BIRATIONAL ALGEBRAIC TOPOLOGY

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Abstract. Over a qcqs scheme S, we analyze the birational localization $L_{\mathrm{bir}}\mathcal{H}^{\mathbb{A}^1}(S)$ of the motivic ∞-category $\mathcal{H}^{\mathbb{A}^1}(S)$. We establish that the associated localization functor L_{bir} commutes with the bar construction, thereby preserving connectivity over fields. When the field is perfect, we show that a sheaf of groups is birational exactly when it is strongly $\mathbb{A}^1$-invariant and has a trivial $\mathbb{G}_m$-contraction. For connected motivic spaces over such fields, this yields a canonical equivalence between L_{bir} and the $S^{2,1}$-nullification functor $L^{2,1}$. For a general field k, we identify $\pi_0^b(X)$ with $\pi_0^{b\mathbb{A}^1}(X)$ for proper k-schemes and show that $\mathbb{A}^1$-connectedness is equivalent to birational connectedness for (ind-)proper k-schemes. This also confirms that $\pi_0^{b\mathbb{A}^1}(-)$ and $\mathbb{H}_0^{\mathbb{A}^1}(-)$ are stable birational invariants of smooth proper k-schemes, satisfying the expected universal property for invariants valued in birational sheaves (of sets and abelian groups respectively). Finally, we show that etale birational equivalences to S are precisely the dense open immersions into S.

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1. Introduction to Birational Algebraic Topology

1.1. **The aims of birational motivic homotopy theory.** Birational geometry studies birational morphisms of schemes. Denoted $X \dashrightarrow Y$, a birational morphism from X to Y stands for a span $X \xleftarrow{i} U \xhookrightarrow{j} Y$ of dense open immersions (i.e., both i and j have homeomorphic images that are dense in their codomain and induce isomorphisms $i^*\mathcal{O}_X \simeq \mathcal{O}_U \simeq j^*\mathcal{O}_Y$ of structure sheaves). One of the classical questions in birational geometry is the rationality problem: How to determine whether a given variety is birational to $\mathbb{A}^n$? However clean the statement may seem, the rationality problem has no universal solution.

The issue is that birational morphisms are not morphisms in any sufficiently well-behaved 1-category that suitably enlarges the category of (smooth) schemes. Moreover, the rationality question is a rigid one, in the sense that it is really about isomorphism classes. Problems of this type often require one to consider the 'possible' $(2,1)$-category of such spans, which extends the 1-category of (smooth) schemes in a suitable manner.

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One of the best techniques for obtaining negative answers to such problems is to construct invariants that induce isomorphisms on such spans. However, an appropriate, well-behaved invariant must satisfy the properties of a good cohomology theory, for which even 2-categories are insufficient. One shall rather build a category that has all higher categorical colimits. The 'minimal' such category, which contains all smooth schemes and is a natural home for such spans, is precisely the point of the birational homotopy category. This is the localization of $\mathcal{P}_\Sigma(Sm_S)$ (see [Lur09, Definition 5.5.8.8] for the notation) at the set of dense open immersions, denoted $\mathcal{H}^b(S) := L_{dense}\mathcal{P}_\Sigma(Sm_S)$.

Frankly, if one really wants cohomology theories of sufficiently good nature, a priori, it is not clear why our birational category should enforce them. Therefore, it is suggestive to start with a good cohomological extension rather than $\mathcal{P}_\Sigma(Sm_S)$. One of the best and most popular such unstable candidates is, no doubt, the $\mathbb{A}^1$ homotopy category, constructed by Morel and Voevodsky [MV99] (recall that this is defined as the $\mathbb{A}^1$-localization of the Nisnevich ∞-topos of Sm_S). So the correct candidate for birational homotopy shall rather be $\mathcal{H}^{b\mathbb{A}^1}(S) := L_{dense}\mathcal{H}^{\mathbb{A}^1}(S)$.

In fact, historically, this was the first construction of the 'birational motivic homotopy category' and appeared in [Pel14] for a completely different reason: to study a possible slice filtration of the unstable motivic homotopy category. This unstable theory has a significant impact on motivic slice computations (see [BE21]). (We will carry out a similar computation in another work [Mai26a]).

With a different goal in mind, Choudhury and Roy [CR24] considered both constructions over a field k and showed in Theorem 3.4 of *loc.cit.* that they coincide. That is, homotopy invariance and Nisnevich locality are automatic in $\mathcal{H}^b(k)$. This result also appeared in a work of Cisinski and Kahn [CK23, Corollary 2.3]. However, both contain a technical oversight: they rely on the unrestricted ∞-category of presheaves $\mathcal{P}(Sm_k)$ rather than $\mathcal{P}_\Sigma(Sm_k)$, so that their $\mathbf{H}_b(k)$ (which evaluates to $L_{dense}\mathcal{P}(Sm_k)$ in our setting) contains objects that fail to be local with respect to disjoint covers. The first section of this paper addresses this gap by providing a rigorously corrected version of this strategy while generalizing it to all base schemes. Thus, we show that for any qcqs scheme S, there is an equality of ∞-categories $\mathcal{H}^{b\mathbb{A}^1}(S) = \mathcal{H}^b(S)$. (The strategy we develop for automatic Nisnevich locality differs from that of [CR24]. In another work [Mai26d, Theorem 2.3.6], I provide a refinement of their Corollary 3.7 by employing the framework of $\mathcal{P}_\Sigma(Sm_S)$, carried out uniformly over all qcqs base schemes.)

Now, obviously, the $\mathbb{A}^1$ homotopy category is not the most general cohomologically rich construction, simply because not all cohomologies are $\mathbb{A}^1$-invariant. There are a few more on the list that do a little better. Among many such candidates, another unstable cohomological category is the logarithmic motivic homotopy category of [BPOsr22]. In §2.3 of this paper, we will see that even with the logarithmic setup, one does not get anything new: the 'strict logarithmic birational motivic' homotopy category is equivalent to the ordinary 'birational motivic homotopy category'.

We may therefore unambiguously denote the birational motivic homotopy category by $\mathcal{H}^b(S)$. It will therefore follow that computations in $\mathcal{H}^{b\mathbb{A}^1}(S)$ are significantly simpler than their $\mathbb{A}^1$-homotopical counterparts.

This tractability brings us to the other important aspect: to what extent can birational problems be resolved entirely via $\mathbb{A}^1$-homotopy theory? This question has been raised and partially answered in various accounts. The starting point of this line of study is probably [AM11], where the authors show that, for smooth proper varieties, the sections of the $\mathbb{A}^1$-homotopy path component over $\mathcal{F}_k$ capture Manin's notion of R-equivalent points. However, it is known that $\pi_0^{\mathbb{A}^1}X$ (even for a smooth proper variety X) is not birational [BHS15, Example 4.8], so we are forced to consider the 'universal birationalization'. But this birationalization turns out to be difficult to compute by hand.

In later explorations [Aso13], [AH11], the authors have used concepts like this, paired with stability/homology, to detect the existence of rational points or zero cycles of degree one. In fact, they conjectured that the zeroth $\mathbb{A}^1$ homology of such varieties is the free abelian sheaf of their $\pi_0^{b\mathbb{A}^1}$ (this has been verified in [Koi22]) and is itself a birational invariant. Connections of these aspects to birational homotopy theory were established in the other initial references [CR24, Theorem 3.5], [CK23, Proposition 1.9]: the unstable birational path component $\pi_0^b X$ is isomorphic to $\pi_0^{b\mathbb{A}^1} X$ for a smooth proper variety X/k.

In this paper, we remove the smoothness hypothesis via an alternative approach [Theorem 3.6.15] and apply it in Corollary 3.6.16 to show that for a proper scheme X/k, $\pi_0^{b\mathbb{A}^1} X$ [AM11] is precisely the aforementioned universal birationalization of $\pi_0^{\mathbb{A}^1} X$ (in fact, inducing an isomorphism over $\mathcal{F}_k$). Furthermore, this establishes that proper motivic spaces are $\mathbb{A}^1$-connected if and only if they are birationally connected [Corollary 3.6.17].

It must, however, be noted that the full motivic homotopy type $L_{mot}X$ of a proper scheme X differs significantly from its birational homotopy type $L_{bir}X$. Birational localization thus yields genuinely new birational invariants that remain invisible to the motivic localization.

(In [Mai26c], we shall explain another important aspect of moving from the $\mathbb{A}^1$-homotopy category to the Birational homotopy category, namely, to add topos theoretic properties to $\mathcal{H}^{\mathbb{A}^1}$.)

1.2. **Notations and terminologies.** We shall freely use the language of ∞-categories without referring to any particular model for the theory. The reader may freely use their preferred model. A standard reference is [Lur09], where the author models these on Kan complexes as ∞-groupoids. With a choice of Grothendieck universes, the terms Large and small will be treated as usual. The term $\mathrm{Map}_{_}(-,-)$ will always denote the ∞-groupoid of morphisms, while $[-,-]_{_}$ will denote the path component $\pi_0\mathrm{Map}_{_}(-,-)$. *Spc* will stand for the ∞-category of spaces. Pr^L shall denote the ∞-category of large presentable ∞-categories, while Cat_∞ will denote the ∞-category of all large ∞-categories.

For an ∞-category $\mathcal{C}$, by $\mathcal{P}(\mathcal{C})$ we mean the category of presheaves of spaces on $\mathcal{C}$, which is the free co-completion of $\mathcal{C}$. If $\mathcal{C}$ has coproducts, by $\mathcal{P}_\Sigma(\mathcal{C})$ we mean the full subcategory of $\mathcal{P}(\mathcal{C})$ consisting of presheaves that turn finite coproducts into products. This is the sifted cocompletion of $\mathcal{C}$, and therefore the corresponding Yoneda embedding $h : \mathcal{C} \to \mathcal{P}_\Sigma(\mathcal{C})$ preserves coproducts [Lur09, Proposition 5.5.8.10]. On the other hand, $\mathcal{P}(\mathcal{C})_\bullet$ will denote the full subcategory of $\mathcal{P}(\mathcal{C})$ of pointed objects. *pt* will stand for the terminal object, so $\mathcal{P}(\mathcal{C})_\bullet = \mathcal{P}(\mathcal{C})_{pt/}$

Throughout this paper, S will denote a scheme, and k will denote a field. In most of the important results, we will need S to be Qcqs and k to be perfect. But we will specify the conditions each time we use them: either at the beginning of the (sub)section or within the statements themselves. We do not assume k is perfect unless otherwise specified. By a morphism of schemes, we shall always mean a separated morphism. Sm_S stands for the category of smooth, finite-type, separated morphisms over S.

We refer to [Hoy14, Appendix C] for the notion of the Nisnevich topology and Nisnevich sheaves on SM_S (smooth schemes over S, not necessarily of finite type) when S is a general scheme. When S is Qcqs, Proposition C.5 (3) of *loc.cit.* yields a canonical equivalence $\mathcal{P}_{nis}(SM_S) \simeq \mathcal{P}_{nis}(Sm_S)$ of the ∞-toposes of Nisnevich sheaves of spaces. Since we will always work with Qcqs schemes, it is fine to work with Sm_S instead of SM_S.

By $\mathcal{P}(S)$, we mean the ∞-category of spaces on Sm_S. Throughout, we let σ denote a Grothendieck topology on Sm_S. In most of the important results, we will need σ to be coarser than (but including) the Nisnevich topology. We denote the topos of σ-local (∞-)sheaves as $\mathcal{P}_\sigma(S)$. The corresponding localization functor, denoted L_σ, is the associated σ (∞)-sheafification and is

hence left exact. π_n^σ denotes the homotopy groups internal to the $\mathcal{P}_\sigma(S)$-local topos, i.e., the σ sheafification of the presheaf π_n.

We shall use the notions of accessible localization and saturation classes throughout the paper. A standard reference is [Lur09, §5.4.4]. We will use the notation $\mathcal{H}^{\mathbb{A}^1}(S)$ for the full subcategory of $\mathcal{P}(S)$ consisting of S-motivic spaces i.e., $\mathbb{A}^1$-local Nisnevich-local objects. The corresponding localization functors will be denoted $L_{\mathbb{A}^1}, L_{nis}, L_{mot}$, etc. $\pi_i^{\mathbb{A}^1}$ will, as usual, stand for $\pi_i^{nis}L_{mot}$. In principle, one may use a combination of a topology σ and a smooth scheme A/S to construct a localization $\mathcal{H}^A_\sigma(S) := L_A\mathcal{P}_\sigma(S)$.

We use the notation PSh_S, $PGrp_S$, and PAb_S for the categories of presheaves on Sm_S with values in sets, groups, and abelian groups, respectively. For a topology σ, we use the notation $Sh^\sigma_S, Grp^\sigma_S, Ab^\sigma_S$ for the corresponding subcategories of σ-sheaves. When $\sigma = nis$, we denote these as usual by Sh_S, Grp_S, Ab_S. On the other hand, $Sh_S^{\mathbb{A}^1}, Grp_S^{\mathbb{A}^1}, Ab_S^{\mathbb{A}^1}$ denote the subcategories of ordinary, strongly, and strictly $\mathbb{A}^1$-invariant sheaves, respectively.

When k is a field, $\mathcal{F}_k$ denotes the category of finitely generated separable extensions of k and field homomorphisms. We write Sch_k, Prop_k, and SmProp_k for the categories of all schemes, proper schemes, and smooth proper schemes over k, respectively.

1.3. **Summary of the paper.** Over a general Qcqs base scheme S, we study the algebraic topology of the Birational Motivic homotopy category $L^0_{bir}\mathcal{H}^{\mathbb{A}^1}(S)$ of [BE21]. Recall that this is defined as the localization of the Motivic homotopy category $\mathcal{H}^{\mathbb{A}^1}(S)$ at the set $B_0(S)$ of dense open immersions in Sm_S. To shorten notation and be consistent with the notation of [AM11], we denote this localization by $\mathcal{H}^{b\mathbb{A}^1}(S)$. By dropping 0, we denote the localization functor L^0_{bir} of [BE21] simply by L_{bir}.

The central idea of the paper is that localization at $B_0(S)$ is strong enough to enforce both Nisnevich and $\mathbb{A}^1$-locality, provided one starts with the sifted cocompletion $\mathcal{P}_\Sigma(Sm_S)$. In fact, it is strong enough to induce locality for any rational scheme.

In other words, if we define:

(1) The Birational homotopy category as $\mathcal{H}^b(S) := L_{dense}\mathcal{P}_\Sigma(Sm_S)$,
(2) The Birational Motivic homotopy category as $\mathcal{H}^{b\mathbb{A}^1}(S) := L_{dense}\mathcal{H}^{\mathbb{A}^1}(S)$,
(3) The rational motivic homotopy category $\mathcal{H}^{rat}(S)$ as the localization of $\mathcal{P}_{nis}(Sm_S)$ at all rational schemes over S,

then,

Theorem 1 (see Corollary 2.2.9 and Corollary 2.2.12)**.** $\mathcal{H}^b(S) \subset \mathcal{H}^{rat}(S) \subset \mathcal{H}^{\mathbb{A}^1}(S)$. *In particular,* $\mathcal{H}^{b\mathbb{A}^1}(S) \simeq \mathcal{H}^b(S)$.

In the same section, we define two analogs of the birational localization for the logarithmic motivic homotopy category log-$\mathcal{H}^\square(S)$ [BPOsr22]:

(1) log-$\mathcal{H}^{log\text{-}b}(S)$: the localization of log-$\mathcal{H}^\square(S)$ at all log-dense open immersions of log schemes,
(2) log-$\mathcal{H}^b(S)$: the localization of log-$\mathcal{H}^\square(S)$ at all dense open immersions of log schemes.

We then show that:

Theorem 2 (see Corollary 2.3.6)**.** *There is a canonical equivalence:* log-$\mathcal{H}^b(S) \simeq \mathcal{H}^b(S)$ *induced by the forgetful functor* $\rho : SmlSm_S \to Sm_S$ *given by* $(X, D) \mapsto X$.

By choosing to work with Qcqs schemes, we shall therefore consider $\mathcal{H}^b := L_{dense}\mathcal{P}_\Sigma$ as the candidate for the birational motivic homotopy category. This simplicity of the birational motivic homotopy category will make the corresponding algebraic topology extremely simple. For example, we show that:

Theorem 3 (see Lemma 3.2.3)**.** *L_{bir} commutes with the two-sided (Nisnevich) bar construction.*

One must be cautioned that similar statements are incorrect for the motivic localization functor L_{mot}, because the inclusion $\mathcal{H}^{\mathbb{A}^1}(S) \subset \mathcal{P}(S)$ is not closed under geometric realization. The problem stems from a similar issue with the inclusion $\mathcal{P}_{nis}(S) \subset \mathcal{P}(S)$. Although this last inclusion (and hence the other one) is closed under filtered colimits (because of the cd nature of the Nisnevich topology), it is not closed under sifted colimits. This is precisely the reason Morel's notion of Strong/Strict $\mathbb{A}^1$-invariance arises, which makes ordinary motivic homotopy theory more complicated.

In fact, a higher m connectivity theorem for L_{bir} follows immediately from the same facts, which, in the case of $\mathcal{H}^{\mathbb{A}^1}$, is highly sensitive to the dimension of the base scheme:

Theorem 4 (see Corollary 3.5.9)**.** *Let S be the spectrum of a henselian local ring. Then, L_{bir} takes Nisnevich locally n-connected (in fact, $\mathbb{A}^1$-n-connected) S-spaces to sectionwise n-connected S-spaces.*

It turns out that commutation of L_{bir} with the bar construction is sufficient to import many more topological results directly from the parent topos. Some of these results also hold in the motivic case, though they require more effort, and having a singular model of $L_{\mathbb{A}^1}$ saves most of them.

Unfortunately, in the birational case, we are unable to construct an explicit formula for L_{bir} in general. However, over perfect fields, we will prove that this is possible, at least for connected spaces. The key observation leading to this model is the following theorem:

Theorem 5 (see Theorem 3.3.12)**.** *Let k be a perfect field. Then the following conditions for a Nisnevich-locally connected pointed motivic space $\mathcal{X}$ over k are equivalent:*

(1) $\mathcal{X}$ is birational local.
(2) $\pi_i^{nis}\mathcal{X}$ is birational local for all i.
(3) $(\pi_i^{nis}\mathcal{X})_{-1} = 0$ for all i.
(4) $\mathcal{X}$ is $\mathbb{P}^1$ local (equivalently $S^{2,1}$ null [ABH23]).

In other words, $\tau_{\geq 1}^{nis}\mathcal{H}^b(k) = \tau_{\geq 1}^{nis}(\mathcal{H}^{\mathbb{A}^1}(k) \bigcap \mathcal{H}^{\mathbb{P}^1}(k))$.

We shall, however, point out that this theorem might fail for non-connected spaces, as Counterexample 3.3.15 shows. Nevertheless, it follows immediately that when k is a perfect field, there is an equivalence of functors

$$L_{bir}\big|_{\tau_{\geq 1}^{nis}\mathcal{H}^{\mathbb{A}^1}(k)} \simeq L^{2,1}\big|_{\tau_{\geq 1}^{nis}\mathcal{H}^{\mathbb{A}^1}(k)}$$

where $L^{2,1}$ denotes the $S^{2,1}$ nullification from [ABH23]. Although the authors of *loc.cit.* provide a formula for $L^{2,1}$, we caution that it is obtained only via a small object argument. To complete this modeling of L_{bir}, we will therefore provide explicit models for $L^{2,1}$ localization using a 'singular-like' construction [Theorem 3.4.2].

In the final subsection, we will obtain standard results about connected coverings (see Theorem 3.6.7, Proposition 3.6.8) and then show that:

Theorem 6 (see Corollary 3.6.12)**.** *Suppose $\mathcal{X}$ is a (motivically) pointed $\mathbb{A}^1$-n-connected S-space. Then the canonical map $\pi^{\mathbb{A}^1}_{n+1}\mathcal{X} \to \pi^b_{n+1}\mathcal{X}$ is universal among maps to a birational sheaf of sets when $n = -1$; to a birational sheaf of groups when $n = 0$; and to a birational sheaf of abelian groups when $n \geq 1$; i.e., $\pi^b_{n+1}\mathcal{X} \simeq \pi^b_0(\pi^{\mathbb{A}^1}_{n+1}\mathcal{X})$.*

In fact, using Proposition 3.6.8, we will obtain a generalization of [CR24, Theorem 3.5] to all proper schemes (see [AM11, Theorem 6.2.1] for the definition of $\pi_0^{b\mathbb{A}^1}X$):

Theorem 7 (see Theorem 3.6.15, Corollary 3.6.16, and Corollary 3.6.17)**.** *Let k be a field and X a proper scheme over k. Then there is a canonical isomorphism $\pi^b_0X \simeq \pi_0^{b\mathbb{A}^1}X$. It follows that the canonical morphism $\pi_0^{\mathbb{A}^1}X \to \pi_0^{b\mathbb{A}^1}X$ is the universal birationalization and that $\pi_0^{b\mathbb{A}^1}(-)$ is a stable birational invariant of smooth proper schemes.*

As a consequence, we obtain:

Theorem 8 (see Corollary 3.6.18, Corollary 3.6.19)**.** *For smooth proper schemes, $X \to \mathbb{H}_0^{\mathbb{A}^1}X$ is the universal birational sheaf of abelian groups, and the assignment $X \mapsto \mathbb{H}_0^{\mathbb{A}^1}X$ is a stable birational invariant.*

Theorem 9 (see Corollary 3.6.22)**.** *An (ind-) proper scheme over a field k is $\mathbb{A}^1$-connected if and only if it is birationally connected.*

As a final application of Proposition 3.6.8, we prove a birational version of the Freudenthal suspension theorem (see Theorem 3.6.26).

We will conclude the paper in §4 with a collection of examples of birational motivic spaces and birational motivic equivalences. In the final theorem of this section, we characterize dense open immersions among birational equivalences:

Theorem 10 (see Corollary 4.2.40)**.** *If S is a Qcqs scheme, then a morphism $f : X \to S$ is a dense open immersion iff it is étale and a birational equivalence over S.*

2. Constructing the Birational Motivic Homotopy Category

Let S be a Qcqs scheme. By Sm_S we mean the full subcategory of Sch_S spanned by schemes over S with finitely presented and smooth structure maps. By an S-space we mean a presheaf of spaces on Sm_S. The ∞-category of S-spaces, denoted $\mathcal{P}(S)$, is defined as the ∞-category of presheaves of spaces on Sm_S, i.e.,

$$\mathcal{P}(S) := \mathcal{P}(Sm_S) = \mathrm{Fun}(Sm_S^{op}, Spc).$$

Since Sm_S is (essentially) small, by [Lur09, Remark 5.5.3.7] the ∞-category $\mathcal{P}(S)$ is presentable and, in fact, an ∞-topos (see [Lur09, Definition 6.1.0.4]). When the base scheme S is fixed, we refer to S-spaces simply as 'spaces' and implicitly identify any smooth scheme X with its representable presheaf $h_S(X)$.

We say that an open immersion (see [Sta26, Definition 01HE]) $j : U \hookrightarrow X$ is a dense open immersion if the underlying topological space $j(U)$ is dense in X. We say that a presheaf $\mathcal{X} \in \mathcal{P}(S)$ is dense local if it is local with respect to the class

$$B_0(S) := \{j : U \hookrightarrow X \in Sm_S \mid j \text{ is a dense open immersion}\}$$

of dense open immersions in Sm_S. We denote the full subcategory of $\mathcal{P}(S)$ consisting of dense local spaces by $L_{dense}\mathcal{P}(S)$. Moreover, since $B_0(S)$ is (essentially) small, it follows from [Lur09, Proposition 5.5.4.15] that $L_{dense}\mathcal{P}(S)$ is an accessible localization of $\mathcal{P}(Sm_S)$. Let us denote the associated saturated class and the corresponding localization functor by $Dense(S)$ and L_{dense}, respectively.

It is well known that the Yoneda embedding $h : Sm_S \to \mathcal{P}(Sm_S)$ does not, in general, preserve coproducts. Consequently, the ∞-category $\mathcal{P}(\mathcal{S}) := \mathcal{P}(Sm_S)$ contains many (geometrically) unidentified homotopy types. As a result, the dense local homotopy category $L_{dense}\mathcal{P}(S)$ is too naive to support meaningful birational geometry. To avoid introducing new homotopy types for existing coproducts, one should replace $\mathcal{P}(Sm_S)$ with the category $\mathcal{P}_\Sigma(Sm_S)$ [Lur09, Definition 5.5.8.8]. This is the full subcategory of $\mathcal{P}(Sm_S)$ consisting of presheaves that send finite coproducts in Sm_S to products in Spc.

Remark 2.0.1. Since Sm_S is an extensive (∞-)category (i.e., has disjoint coproducts), the category $\mathcal{P}_\Sigma(S) := \mathcal{P}_\Sigma(Sm_S)$ is equivalent to the ∞-category $\mathcal{P}_\sqcup(Sm_S)$ of sheaves of spaces on Sm_S with the disjoint cover topology (i.e., $\{U_\alpha \to X\}$ is a cover iff $\sqcup U_\alpha \to X$ is an isomorphism) [BH21, Lemma 2.4]. Thus, the localization functor

$$L_\Sigma : \mathcal{P}(Sm_S) \to \mathcal{P}_\Sigma(Sm_S)$$

is left exact and accessible, making $\mathcal{P}_\Sigma(Sm_S)$ an ∞-topos. Clearly, the Nisnevich topology is finer than the $\sqcup$-topology and hence $\mathcal{P}_{nis}(S) \subset \mathcal{P}_\Sigma(S)$.

With this understanding, we are now ready to define the correct version of the birational homotopy category:

Definition 2.0.2. *A presheaf $\mathcal{X} \in \mathcal{P}(S)$ is said to be birational local if it belongs to $\mathcal{P}_\Sigma(S)$ and is dense local. The full subcategory of $\mathcal{P}(S)$ consisting of birational local spaces is called the* ***Birational Homotopy category****, denoted $\mathcal{H}^b(S)$. Since $B_0(S)$ is an essentially small set, we note that $\mathcal{H}^b(S) \subset \mathcal{P}(S)$ is an accessible localization. The associated saturated class is denoted $bir(S)$, and a morphism in it is called a Birational equivalence.*

In fact, one may strengthen the geometric identification by imposing descent for a finer topology σ and, analogously, define $\mathcal{H}^b_\sigma(S)$ as the localization of $\mathcal{P}_\sigma(Sm_S)$ at the set $B_0(S)$. Let us denote the corresponding saturated class by $bir_\sigma(S)$. If σ is finer than τ, then $\mathcal{H}^b_\sigma \subset \mathcal{H}^b_\tau$. In particular, we have

$$\mathcal{H}^b_{nis} \subset \mathcal{H}^b_\sqcup \equiv \mathcal{H}^b.$$

On the other hand, to obtain a cohomologically rich homotopy category that also respects birationality, one may as well start with the ∞-category $\mathcal{H}^{\mathbb{A}^1}(S)$ of S-Motivic spaces from [MV99] rather than with $\mathcal{P}_{nis}(S)$. Recall that a space $\mathcal{X} \in \mathcal{P}(S)$ is an S-Motivic space if it is a nisnevich sheaf and local with respect to the projection maps

$$\mathbb{A}(S) := \{\mathbb{A}^1_X \to X\}_{X/S \in Sm_S}.$$

We define a cohomological analog of the birational category as the localization of the motivic ∞-category $\mathcal{H}^{\mathbb{A}^1}(S)$ at $B_0(S)$. Let us denote this new category by $\mathcal{H}^{b\mathbb{A}^1}(S)$. Again, since $B_0(S)$ is (essentially) small, this is the full subcategory of $\mathcal{H}^{\mathbb{A}^1}(S)$ consisting of dense local objects, i.e.,

$$\mathcal{H}^{b\mathbb{A}^1}(S) := L_{dense}\mathcal{H}^{\mathbb{A}^1}(S).$$

This is the 0-birational motivic homotopy category $L^0_{bir}\mathcal{H}^{\mathbb{A}^1}(S)$ of [BE21] (indeed, it is easy to see that 0-birational maps, as defined in *loc.cit.*, are precisely the dense open immersions). We will thus drop the 0 everywhere and call the category

$$\mathcal{H}^{b\mathbb{A}^1}(S) \equiv L^0_{bir}\mathcal{H}^{\mathbb{A}^1}(S)$$

the Birational **Motivic** homotopy category, and denote the associated localization functor L^0_{bir} from *loc.cit.* simply by L_{bir}. By construction, there is a canonical natural transformation $L_{mot} \to L_{bir}$ that is an L_{bir}-equivalence.

Remark 2.0.3. The dense localization of the motivic homotopy category is a strictly non-trivial localization. For example, it identifies the $\mathbb{A}^1$-local scheme $\mathbb{G}_m$ with the $\mathbb{A}^1$-contractible scheme $\mathbb{A}^1$ by the dense open immersion $\mathbb{G}_m \hookrightarrow \mathbb{A}^1$.

Since $\mathcal{H}^{\mathbb{A}^1}(S)$ and $\mathcal{P}_\sigma(S)$ are presentable and $B_0(S)$ is essentially small, by [Lur09, Proposition 5.5.4.15], each member of the chain of inclusions

$$\mathcal{H}^{b\mathbb{A}^1} \subset \mathcal{H}^b_{nis} \subset \mathcal{H}^b_{\sqcup}$$

is a presentable ∞-category. The aim of the next two subsections is to show that each of these inclusions is an equality.

2.1. **Nisnevich locality of Birational local spaces.** In this section, we aim to show that $\mathcal{H}^b(S) \subset \mathcal{P}_{nis}(S)$. That is, a birational local space, in the sense of Definition 2.0.2, is automatically Nisnevich local. Equivalently, all Nisnevich local equivalences are birational equivalences. In other words,

$$\mathcal{H}^b(S) := L_{dense}\mathcal{P}_\Sigma(S) = L_{dense}\mathcal{P}_{nis}(S) =: \mathcal{H}^b_{nis}(S).$$

Before we do that, let us note something preliminary:

Proposition 2.1.1. *Let $r : S^{red} \to S$ be the reduction map. Then $r^* : Et_S \to Et_{S^{red}}$ is an equivalence of categories.*

Proof. Since r is a homeomorphism, this follows from [Gro66, Théorème (18.1.2)]. □

Lemma 2.1.2. *Let X/S be smooth. Then, for a dense open immersion $V \hookrightarrow Y$ of schemes smooth over S, the pullback morphism $X \times_S V \to X \times_S Y$ is also a dense open immersion over S.*

Proof. Openness is standard. For denseness, let $W \subset X \times_S Y$ be an open set. We have to show that $W \cap (X \times_S V) \neq \emptyset$. Since smooth maps are (universally) open, the image $f(W)$ of W under the smooth map $f : X \times_S Y \to Y$ is open in Y and hence intersects the dense open subset V of Y, say at $y \in V \cap f(W)$. A preimage $t \in W$ of y must therefore be in $X \times_S V$. □

Here is the main theorem of this subsection:

Theorem 2.1.3. *For any Qcqs scheme S, one has $\mathcal{H}^b(S) \subset \mathcal{P}_{nis}(S)$. Therefore, the inclusion $\mathcal{H}^b_{nis}(S) \subset \mathcal{H}^b_{\sqcup}(S) \equiv \mathcal{H}^b(S)$ is an equality.*

Proof. Since our base scheme is Qcqs (i.e., coherent), by [Hoy14, Proposition C.5 (3), (4)], a presheaf on Sm_S is a Nisnevich sheaf if and only if it satisfies Nisnevich excision. By definition, $\mathcal{X} \in \mathcal{P}(Sm_S)$ satisfies Nisnevich excision if it is contractible over the empty scheme and takes Nisnevich cd squares to pullback squares of spaces. So let $\mathcal{X}$ be a birational local space, i.e., $\mathcal{X} \in L_{dense}L_\Sigma\mathcal{P}(Sm_S)$. Since $\mathcal{X} \in \mathcal{P}_\Sigma(Sm_S)$, we have $\mathcal{X}(\emptyset) \simeq pt$ for free. For the excision part, consider a Nisnevich cd square:

$$\begin{array}{ccc} V & \hookrightarrow & Y \\ {\scriptstyle q}\downarrow & & \downarrow{\scriptstyle p} \\ U & \underset{j}{\hookrightarrow} & X \end{array}$$

where j is an open subscheme and p is etale. Consider the open subscheme $W = X \setminus \bar{U}$. Clearly, $U \sqcup W \hookrightarrow X$ is a dense open immersion. Since $p_{|Y\setminus V} : Y \setminus V \to X \setminus U$ is an isomorphism of reduced schemes and $W \subset X \setminus U$, the pullback

$$p^{red}_{|p^{-1}W} : p^{-1}W^{red} \to W^{red}$$

is an isomorphism. Since both W and $p^{-1}W$ are etale over X, by Proposition 2.1.1 we deduce that the original map $p_{|p^{-1}W} : p^{-1}W \to W$ is also an isomorphism. On the other hand, since $U \sqcup W \hookrightarrow X$ is dense, by Lemma 2.1.2 so is $V \sqcup p^{-1}W \hookrightarrow Y$. Now, let us consider the following commutative diagram:

$$\begin{array}{ccccccc}
\mathcal{X}(U) \times \mathcal{X}(W) & \xleftarrow[\simeq]{} & \mathcal{X}(U \sqcup W) & \xleftarrow[\simeq]{} & \mathcal{X}(X) & \longrightarrow & \mathcal{X}(U) \\
\downarrow{\scriptstyle q^* \times p^*_{|p^{-1}W}} & & & & \downarrow & & \downarrow{\scriptstyle q^*} \\
\mathcal{X}(V) \times \mathcal{X}(p^{-1}W) & \xleftarrow[\simeq]{} & \mathcal{X}(V \sqcup p^{-1}W) & \xleftarrow[\simeq]{} & \mathcal{X}(Y) & \longrightarrow & \mathcal{X}(V)
\end{array}$$

The horizontal arrows on the left are equivalences because $\mathcal{X} \in L_\Sigma \mathcal{P}(Sm_S)$. The horizontal arrows in the middle are equivalences because $\mathcal{X}$ inverts dense open immersions. Since $p_{|p^{-1}W}$ is an isomorphism, the entire outer diagram (with the left and middle horizontal arrows replaced by their inverses) is tautologically a pullback square. Thus, the square on the extreme right (being equivalent to the whole outer diagram) is a pullback square itself. □

Remark 2.1.4. Alternatively, one can directly show that Nisnevich-local equivalences over Qcqs schemes are birational equivalences in our sense. This is established in [Mai26d, Theorem 2.3.5]. The argument there rectifies a slight inaccuracy in the formulation of the analogous result over fields, proven in [CR24, Lemma 3.6]. Because the framework in [CR24] relies on the naive dense local homotopy category (i.e., their birational category $\mathbf{H}_b(k)$ is merely $L_{dense}\mathcal{P}(k)$), the *birational weak equivalences* they consider are, in fact, just the dense local equivalences in our sense. However, for a nontrivial decomposition $A \sqcup B = X$ (which is a Nisnevich cover) of smooth schemes, the Čech nerve of the presheaf map

$$h_S A \sqcup h_S B \to h_S X$$

need neither be a sectionwise equivalence nor a dense local equivalence. In [Mai26d, Theorem 2.3.4] we address this discrepancy by providing a modified proof while simultaneously working over an arbitrary Qcqs base scheme.

Remark 2.1.5. Using similar techniques, one can show that birational local spaces are also (smooth) cdp local. Therefore, birational local spaces are scdh local.

We conclude this section with a discussion of the implications of the above theorem. First, we present a generalization of a well-known definition (see [AM11, Definition 6.1.1]):

Definition 2.1.6. *A presheaf of sets F on Sm_S is called dense local if it is $B_0(S)$-local in $PSh(S)$. We call it birational if, in addition, F lies in $PSh_\Sigma(S)$. In other words, these are precisely the discrete spaces that are birational local.*

Remark 2.1.7. From Theorem 2.1.3, it follows that, over any Qcqs scheme S, every birational presheaf is automatically a Nisnevich sheaf. We will denote by Shv^b_S, Grp^b_S, and Ab^b_S the categories of birational (pre)sheaves on Sm_S with values in sets, groups, and abelian groups, respectively.

Proposition 2.1.8. *A presheaf $\mathcal{X}$ of spaces is birational-local iff, for every $i \geq 0$, the presheaf $\pi_i\mathcal{X}$ is birational.*

Proof. It is clear that $\mathcal{X}$ is dense local iff $\pi_i\mathcal{X}$ is dense local for all i. Thus it suffices to know that $\mathcal{X} \in \mathcal{P}_\Sigma(Sm_S)$ iff $\pi_i\mathcal{X} \in PSh_\Sigma(S)$. This is obvious because π_i preserves finite products. □

It follows that when $\mathcal{X}$ is birational local, the presheaves $\pi_i\mathcal{X}$ are birational presheaves of sets for $i = 0$, of groups for $i = 1$, and of abelian groups for $i \geq 2$. By the remark above, Remark 2.1.7,

these are automatically Nisnevich local. Thus, in particular, the birational analog of Morel's $\mathbb{A}^1$-homotopical result (that higher $\mathbb{A}^1$-homotopy group sheaves are $\mathbb{A}^1$-invariant) is straightforward and is true even for the Nisnevich path-component:

Corollary 2.1.9. *Let $\mathcal{X}$ be a birational local space over a Qcqs scheme S and let σ be a topology coarser than the Nisnevich topology. Then for all $i \geq 0$, $\pi_i^\sigma \mathcal{X} = \pi_i \mathcal{X}$. It follows that the sheaves $\pi_i^\sigma \mathcal{X}$ are birational local. In particular, the Nisnevich sheaf of sets $\pi_i^{nis}\mathcal{X}$ is birational for all $i \geq 0$.*

Proof. By Proposition 2.1.8, we know that $\pi_i\mathcal{X}$ is birational. Using Theorem 2.1.3, we therefore have that $\pi_i\mathcal{X}$ is Nisnevich local and thus σ-local, for all σ coarser than the Nisnevich topology. Consequently, $\pi_i\mathcal{X} = \pi_i^\sigma\mathcal{X}$. Because $\pi_i\mathcal{X}$ is birational, in turn so are $\pi_i^\sigma\mathcal{X}$. □

Remark 2.1.10. An analogous result for $\mathcal{H}^{\mathbb{A}^1}(S)$ is false: π_0^{nis} of an S-motivic space need not be S-motivic. Until the last decade, it was a well-studied conjecture of Morel [Mor12, Conjecture 1.12] and was recently proven to be false by Ayoub [Ayo23].

2.2. **Rationality of Birational local spaces.** The purpose of this section is to show that Birational local spaces are *Rational local*. In particular, $\mathcal{H}^b(S) \subset \mathcal{H}^{\mathbb{A}^1}(S)$. In fact, we will prove a stronger result that $L_{dense}\mathcal{P}(Sm_S) \subset L_A\mathcal{P}(Sm_S)$, for any rational scheme $A \in Sm_S$.

Here, by a rational local space, we mean a space that is local with respect to every rational scheme A/S. Recall that a scheme A/S is said to be rational if, for some n, there is a span of dense open immersions $A \hookleftarrow U \hookrightarrow \mathbb{A}^n_S$ over S. The following definition is analogous to the construction of the $\mathbb{A}^1$-motivic homotopy category:

Definition 2.2.1 (A-motivic spaces)**.** *Let $A \in Sm_S$. We say that a presheaf $\mathcal{X}$ of spaces over S is A-local if the canonical map $\mathcal{X}(X) \to \mathcal{X}(A \times_S X)$ induced by the projection $A \times_S X \to X$ (where $X \in Sm_S$ is arbitrary) is an equivalence of ∞-groupoids. Denote by $L_A\mathcal{P}(S)$ the full subcategory of A-local spaces. We say that $\mathcal{X}$ is an A-motivic space if, furthermore, it is Nisnevich local. We denote the full subcategory of A-motivic spaces by $\mathcal{H}^A(S)$.*

Example 2.2.2. The primary example is $\mathcal{H}^{\mathbb{A}^1}(S)$, introduced and studied in [MV99]. For $A = \mathbb{P}^1_S$, this is the projective motivic homotopy category, introduced in [BPOsr22]. We will conduct an extensive study of this second candidate in an upcoming work [Mai26b].

Definition 2.2.3 (Rational Motivic spaces)**.** *We say that a presheaf $\mathcal{X}$ of spaces over S is a rational local space (resp. a rational motivic space) if it is an A-local space (resp. A-motivic space) for every rational scheme $A \in Sm_S$. We denote the full subcategory of rational motivic spaces by $\mathcal{H}^{rat}(S)$.*

Remark 2.2.4. Note that, since Sm_S is (essentially) small, both the inclusions $\mathcal{H}^A(S) \subset \mathcal{P}(S)$ and $\mathcal{H}^{rat}(S) \subset \mathcal{P}(S)$ are accessible localizations, and the resulting categories of local objects are presentable ∞-categories [Lur09, Proposition 5.5.4.15]. We call the associated local equivalences A-motivic equivalences and rational motivic equivalences, respectively.

Remark 2.2.5. Rational local spaces are in particular $\mathbb{A}^1$ local, and thus $\mathcal{H}^{rat}(S) \subset \mathcal{H}^{\mathbb{A}^1}(S)$.

Here is the key result relating birationality to rationality:

Lemma 2.2.6. *Let S be a scheme and $A \in Sm_S$ a rational scheme over S. Then a dense local presheaf of sets over S is A-local.*

Proof. Let F be one such presheaf. We first observe that F is $\mathbb{P}^1$-local, for which one may follow the arguments in [KS15, Appendix A] verbatim. (The assumption in *loc.cit.* that the base is a field is not necessary. See [Mai26d, Lemma 2.3.8] for a quick sketch.)

Since F is dense local, the inclusion $\mathbb{A}^1_S \hookrightarrow \mathbb{P}^1$ as the complement of ∞, induces a natural isomorphism $F(\mathbb{P}^1_S \times_S -) \xrightarrow{\infty} F(\mathbb{A}^1_S \times_S -)$. The previous paragraph then shows that F is $\mathbb{A}^1$-local and hence, for all n, $\mathbb{A}^n$-local as well. Finally, since dense open immersions are stable under base change by smooth morphisms (see Lemma 2.1.2), the span $A \hookleftarrow U \hookrightarrow \mathbb{A}^n$ of dense open immersions induces a natural isomorphism $F(A \times_S -) \to F(\mathbb{A}^n_S \times_S -)$ (again, since F is dense local). By the $\mathbb{A}^n$-locality of F, this immediately implies the A-locality of F. □

Corollary 2.2.7. *Over a Qcqs scheme, birational presheaves are $\mathbb{A}^1$-invariant Nisnevich sheaves.*

Proof. By Lemma 2.2.6, such presheaves are $\mathbb{A}^1$-invariant. These are Nisnevich sheaves by Theorem 2.1.3. □

Lemma 2.2.8. *For any rational scheme A/S we have $L_{dense}\mathcal{P}(S) \subset L_A\mathcal{P}(S)$.*

Proof. A space $\mathcal{E} \in \mathcal{P}(Sm_S)$ is A-local (resp. dense local) if and only if the presheaves $\pi_i(\mathcal{E})$ are A-local (resp. dense local). Therefore, the claim follows immediately from Lemma 2.2.6. □

Corollary 2.2.9. *Let S be a Qcqs scheme. Then:*

(1) For every rational scheme A/S we have,

$$\mathcal{H}^b(S) = \mathcal{H}^b_{nis}(S) \subset \mathcal{H}^A(S).$$

Equivalently, A-motivic equivalences are birational equivalences.

(2) $\mathcal{H}^b(S) \subset \mathcal{H}^{rat}(S)$. Equivalently, rational motivic equivalences are birational equivalences.

Proof. The inclusion in (1) follows immediately from Lemma 2.2.8. Indeed,

$$\mathcal{H}^b_{nis}(S) := \mathcal{P}_{nis}(S) \bigcap L_{dense}\mathcal{P}(S),$$

so that Lemma 2.2.8 yields,

$$\mathcal{H}^b_{nis}(S) \subset \mathcal{P}_{nis}(S) \bigcap L_A\mathcal{P}(S) = \mathcal{H}^A(S).$$

For the equality in 1, see Theorem 2.1.3.

2. The inclusion in (2) follows immediately from the inclusion in 1 and the definition: $\mathcal{H}^{rat}(S) = \bigcap_{A \in Rat_S} \mathcal{H}^A(S)$ (where Rat_S is the set of all rational schemes in Sm_S).

Finally, the 'equivalently' part of each statement is simply the dual of the corresponding inclusion. Indeed, in a presentable ∞-category $\mathcal{C}$, if S_1 and S_2 are two saturated classes, then $\mathrm{S}_1 \subset \mathrm{S}_2$ if and only if $L_{\mathrm{S}_2}\mathcal{C} \subset L_{\mathrm{S}_1}\mathcal{C}$. □

Remark 2.2.10. The birational localization is strictly stronger than the rational localization. Indeed, there are nonrational schemes that are birationally contractible. For example, when $S = Speck$, there is a stably rational smooth k-variety that is not rational [AM11, Example 2.3.4]. But stably rational varieties are clearly birationally contractible.

Remark 2.2.11. One can similarly define $\mathcal{H}^{st\text{-}rat}$, $\mathcal{H}^{ret\text{-}rat}$, etc., as the localization of the Nisnevich topos at all stably rational schemes, retract rational schemes, etc. respectively and similarly the saturated classes *st-rat*, *ret-rat* etc. It is evident that:

$$\mathcal{H}^{bir} \subset \mathcal{H}^{ret\text{-}rat} \subset \mathcal{H}^{st\text{-}rat} \subset \mathcal{H}^{rat} \subset \mathcal{H}^{\mathbb{A}^1}.$$

Dually,

$$mot \subset rat \subset st\text{-}rat \subset ret\text{-}rat \subset bir.$$

Corollary 2.2.12. *Let S be a Qcqs scheme. Then,*

(1) $\mathcal{H}^b(S) \subset \mathcal{H}^{\mathbb{A}^1}(S)$, or, equivalently, motivic equivalences are birational equivalences.
(2) $\mathcal{H}^{b\mathbb{A}^1}(S) = \mathcal{H}^b(S)$.

Proof. Statement (1) follows immediately from Corollary 2.2.9(1) applied to $A = \mathbb{A}^1_S$. Statement (2) can then be deduced easily from 1. Indeed,

$$\mathcal{H}^{b\mathbb{A}^1}(S) = \mathcal{H}^{\mathbb{A}^1}(S) \bigcap L_{dense}\mathcal{P}(S) = \mathcal{H}^{\mathbb{A}^1}(S) \bigcap \mathcal{P}_\Sigma(S) \bigcap L_{dense}\mathcal{P}(S) = \mathcal{H}^{\mathbb{A}^1}(S) \bigcap \mathcal{H}^b(S)$$

(the second equality comes from the fact that $\mathcal{H}^{\mathbb{A}^1}(S) \subset \mathcal{P}_{nis}(S) \subset \mathcal{P}_\Sigma(S)$). □

Remark 2.2.13. A version of the above corollary appears in [CR24, Theorem 3.9] and [CK23, Theorem 2.3]. In these sources, however, the birational homotopy category of a field k (denoted by $\mathbf{H}_b(k)$ and $\mathscr{H}_b(k)$, respectively) is taken simply to be the naive dense local category $L_{dense}\mathcal{P}(k)$. As detailed in Remark 2.1.4, this definition is technically incomplete for the purpose of establishing the inclusion $\mathbf{H}_b(S) \subset \mathcal{P}_{nis}(S)$, and thereby the equivalence $\mathbf{H}_b(S) = \mathcal{H}^{b\mathbb{A}^1}(S)$. The obstruction lies in the fact that objects in $L_{dense}\mathcal{P}(k)$ do not generally satisfy descent with respect to Nisnevich covers given by disjoint union decompositions. For example, the empty presheaf is clearly dense-local, but it is not a Nisnevich sheaf.

Remark 2.2.14. While $\mathbb{A}^1_S \to S$ is a dense equivalence over S, a dense open immersion into S can never be a motivic equivalence. Indeed, suppose $j : U \hookrightarrow S$ is a (dense) open immersion that is a motivic equivalence. Since j is an etale morphism to the base, being a motivic equivalence implies that it is an isomorphism [DVO26, Proposition 3.1].

2.3. **Logarithmic Birational motives.** The logarithmic version of motivic homotopy theory has been extensively studied by Binda, Park, and Østvare in a series of papers starting with [BPOsr22], with the goal of representing more cohomology theories. In fact, this approach has been highly successful in representing even topological Hochschild homology, which is known to be $\mathbb{A}^1$-homotopically contractible.

In the previous subsection, we saw that $\mathcal{H}^b(S)$ is a full subcategory of $\mathcal{H}^{\mathbb{A}^1}(S)$. In fact, using Corollary 2.2.9(1) (applied to the rational schemes $\mathbb{P}^n$), we see that $\mathcal{H}^b(S)$ is also a full subcategory of $\mathcal{H}^{\mathbb{P}^1}(S)$ and $\mathcal{H}^{\mathbb{P}^\bullet}(S)$ (see [BPOsr22, Definition 4.0.3], for notation). This suggests a connection between the birational homotopy category and the logarithmic homotopy category. In this subsection, we will examine this connection. Below, we will thus assume that S is a Qcqs scheme; the reader is free to assume that S is the spectrum of a field.

Consider the forgetful functor $\rho : SmlSm_S \to Sm_S$ defined by $\rho(\bar{X}, D) := \bar{X}$. This has a canonical right adjoint $\lambda : Sm_S \to SmlSm_S$ defined by $\lambda(X) := (X, \emptyset)$ (see [BPOsr22, §4]). Equipped with the logarithmic Nisnevich topology, ρ is neither a morphism of sites nor does it preserve the interval object, so we cannot obtain motivic left derived functors for ρ (it does work for λ, though).

Theorem 2.3.1. *The induced functor $\rho_\sharp : \mathcal{P}(SmlSm_S) \to \mathcal{P}(S)$ sends log motivic equivalences to birational equivalences. Hence, it induces an adjunction*

$$L_{bir}\rho_\sharp : log\text{-}\mathcal{H}^\square(S) \leftrightarrows \mathcal{H}^b(S) : \rho^* \simeq \lambda_\sharp.$$

The right adjoint ρ^* is fully faithful, and we have an embedding

$$\rho^* : \mathcal{H}^b(S) \hookrightarrow \log\text{-}\mathcal{H}^\square(S).$$

Proof. Local equivalences in $\mathcal{P}(SmlSm_S)$ given by strict Nisnevich covers are sent by $\rho_\sharp$ to Nisnevich covers of the total schemes, which are birational equivalences by Theorem 2.1.3. Next, recall that the underlying morphism of total schemes of an admissible blow-up [BPOsr22, Definition 2.5.2, Construction 2.5.9] is given by an ordinary blow-up of schemes. Thus $\rho_\sharp$ maps an admissible blow-up to a birational morphism of schemes, which is obviously a birational equivalence. Finally, $\rho(\bar{\square} \times X) = \mathbb{P}^1_X$, so the projection maps $\bar{\square} \times X \to X$ are sent to $\mathbb{P}^1_X \to X$, which are birational local equivalences by Corollary 2.2.12.

For the last statement, recall that, by definition of log morphisms, the right adjoint λ is fully faithful at the level of schemes. Therefore, the right derived functor $\lambda_\sharp \simeq \rho^*$ is fully faithful. □

We have thus obtained an equivalence $L_{bir}\rho_\sharp\rho^* \to 1_{\mathcal{H}^b(S)}$. So the image of ρ^* is precisely the locus where the unit $1_{\log\text{-}\mathcal{H}^b(S)} \to \rho^* L_{bir}\rho_\sharp$ is an equivalence. We want to describe this image of ρ^*. To do this, we introduce the notion of birational locality in the logarithmic setting.

First, let us call a morphism of log schemes $\underline{U} \to \underline{X}$ a dense open immersion if the underlying morphism of total schemes $\bar{U} \to \bar{X}$ is a dense open immersion of schemes and the log structure on $\underline{U}$ is the pullback of that on $\underline{X}$. Denote by $\log\text{-}\mathcal{H}^b(S)$ the localization of $\log\text{-}\mathcal{H}(S)$ with respect to the class of dense open immersions of log schemes.

Lemma 2.3.2. *$\rho_\sharp : \mathcal{P}(SmlSm_S) \to \mathcal{P}(S)$ preserves birational local equivalences. Hence, the adjunction in Theorem 2.3.1 descends to*

$$L_{bir}\rho_\sharp : log\text{-}\mathcal{H}^b(S) \leftrightarrows \mathcal{H}^b(S) : \rho^* \simeq \lambda_\sharp.$$

Proof. This is automatic by the definition of log-dense open immersions and the above theorem, Theorem 2.3.1. □

Lemma 2.3.3. *$\lambda_\sharp : \mathcal{P}(S) \to \mathcal{P}(SmlSm_S)$ preserves birational local equivalences. Hence, we have an equivalence $L_{log-bir}\lambda_\sharp \simeq \lambda_\sharp$. This yields another adjunction $L_{log-bir}\lambda_\sharp \simeq \lambda_\sharp : \mathcal{H}^b(S) \leftrightarrows \log\text{-}\mathcal{H}^b(S) : \lambda^* \simeq \omega_\sharp$.*

Proof. We know that the class of birational local equivalences is generated by Nisnevich squares and dense open immersions (use Theorem 2.1.3), and that the functor λ trivially preserves Nisnevich squares by the definition of the strict Nisnevich topology on log schemes [BPOsr22, Construction 2.5.9]. It remains to handle dense open immersions, but this is automatic by our definition of log dense open immersions. (Note that, in general, λ does not map $\mathbb{A}^1$-equivalences to equivalences; but we are bypassing this issue by the result of the previous section that $\mathbb{A}^1$-equivalences are dense equivalences).

Since, by the lemma above, $\lambda_\sharp$ preserves birational local objects, we conclude that $L_{log-bir}\lambda_\sharp \simeq \lambda_\sharp$ on $\mathcal{H}^b(S)$. □

Corollary 2.3.4. *$\rho^* : \mathcal{H}^b(S) \hookrightarrow \log\text{-}\mathcal{H}^\square(S)$ factors through $\log\text{-}\mathcal{H}^b(S)$.*

Theorem 2.3.5. *The adjunction* $\rho^* \simeq \lambda_\sharp : \mathcal{H}^b(S) \leftrightarrows \log\text{-}\mathcal{H}^b(S) : \lambda^* \simeq \omega_\sharp$ *as above, is an equivalence.*

Proof. As $\lambda_\sharp$ is fully faithful, we need to show that $\lambda_\sharp\omega_\sharp \to 1$ is an equivalence in $\log\text{-}\mathcal{H}^b(S)$. Because both functors have left adjoints even at the birational level, it suffices to check this on generators, namely smooth log schemes over S. Since for any smooth X we know $\lambda\omega(X) = (X \setminus \partial X, \emptyset)$, it suffices to observe that the obvious open immersion $(X \setminus \partial X, \emptyset) \to (X, \partial X)$ is dense by definition. □

The following is an immediate corollary:

Corollary 2.3.6. *A log motivic space* $\mathcal{X} \in log\text{-}\mathcal{H}^\square(S)$ *is in the essential image of* ρ^* *if and only if it is dense local in* $\mathcal{P}(SmlSm_S)$*. Thus,* ρ^* *induces an equivalence* $\mathcal{H}^b(S) \simeq \log\text{-}\mathcal{H}^b(S)$*.*

At this point, we caution the reader that there is a weaker notion of log-dense maps than the one introduced above. Namely, a dense open immersion $(U, D_U) \to (X, D_X)$ of log schemes is called log-dense if, furthermore, $X \setminus D_X \subset U$ (and therefore $X \setminus D_X \subset U \setminus D_U$, since $D_U = D_X \cap U$). Let us denote the localization of $\log\text{-}\mathcal{H}^\square(S)$ at the log-dense open immersions by $\log\text{-}\mathcal{H}^{log\text{-}b}(S)$. Since log-dense morphisms are in particular dense, it follows that $\log\text{-}\mathcal{H}^b(S) \subset \log\text{-}\mathcal{H}^{log\text{-}b}(S)$. This need not be an equivalence. In fact, this is obtained as a localization at the set $int(S)$ of interior immersions:

$$int(S) := \{X^* = (X \setminus D, \emptyset) \to (X, D)\}_{X \in SmlSm_S}$$

Proposition 2.3.7. *The inclusion* $\log\text{-}\mathcal{H}^b(S) \subset \log\text{-}\mathcal{H}^{log\text{-}b}(S)$ *is a localization with respect to the set* $int(S)$ *of interior immersions. Thus,*

$$\log\text{-}\mathcal{H}^b(S) = L_{int}\log\text{-}\mathcal{H}^{log\text{-}b}(S).$$

Proof. For this, it suffices to observe that a dense open immersion of log schemes is a span of log-dense and interior immersions. Indeed, if $\underline{U} \to \underline{X}$ is a dense open immersion of log schemes, then in the following commutative square of log schemes:

$$\begin{array}{ccc} U^* & \longleftarrow & X^* \\ \downarrow & & \downarrow \\ \underline{U} & \longrightarrow & \underline{X} \end{array}$$

the vertical maps are interior equivalences, while the top horizontal map $U^* \leftarrow X^*$ is trivially log-dense. By 2-out-of-3 property, $\underline{U} \to \underline{X}$ is an equivalence in $L_{int}\log\text{-}\mathcal{H}^{log\text{-}b}(S)$. □

Actually, it is not difficult to see that $w_\sharp$ induces an equivalence $L_{int}\log\text{-}\mathcal{H}^\square(S) \simeq \mathcal{H}^{\mathbb{A}^1}(S)$, and we obtain the following commutative square in Cat_∞:

$$\begin{array}{ccccc} \mathcal{H}^b(S) & \xrightarrow{\sim} & L_{int}\log\text{-}\mathcal{H}^{log\text{-}b}(S) & \hookrightarrow & \log\text{-}\mathcal{H}^{log\text{-}b}(S) \\ \downarrow & & \downarrow & & \downarrow \\ \mathcal{H}^{\mathbb{A}^1}(S) & \xrightarrow{\sim} & L_{int}\log\text{-}\mathcal{H}^\square(S) & \hookrightarrow & \log\text{-}\mathcal{H}^\square(S) \end{array}$$

The candidate $\log\text{-}\mathcal{H}^{log\text{-}b}(S)$ should be regarded as the logarithmically correct candidate for the *logarithmic birational motivic homotopy category* and is canonically larger than $\mathcal{H}^b(S)$. We shall study this candidate in future work.

3. Properties of L_{bir} and the Birational Motivic Homotopy Category

Throughout this section, let S be a Qcqs scheme. The purpose of this section is to study the birational motivic homotopy category $\mathcal{H}^{b\mathbb{A}^1}(S)$ of [BE21,Pel14] and the corresponding localization functor $L_{bir} : \mathcal{P}(S) \to \mathcal{P}(S)$. Because this involves localization with respect to two different sets of morphisms, it seems difficult to study. However, as we saw in the previous section, the Birational motivic homotopy category admits a simpler model given by $\mathcal{H}^b(S) := L_{dense}\mathcal{P}_\Sigma(S)$. Consequently, both $\mathcal{H}^{b\mathbb{A}^1}(S)$ and L_{bir} exhibit surprisingly strong structural properties. For ease of exposition, the subsequent results are formulated in terms of the ∞-category $\mathcal{H}^b(S)$; however, it should be understood that these statements fundamentally apply to the more refined birational motivic homotopy category $\mathcal{H}^{b\mathbb{A}^1}(S)$ and are therefore of motivic significance.

While the Nisnevich topology remains our primary focus, we formulate the following results in the general framework of an arbitrary Grothendieck topology σ on Sm_S. Similarly, given a scheme $A \in Sm_S$, we use a general version of $\mathcal{H}^A(S)$, namely $\mathcal{H}^A_\sigma(S) := L_A\mathcal{P}_\sigma(S)$. This recovers $\mathcal{H}^A(S)$ when $\sigma = nis$ and $\mathcal{P}_\sigma(S)$ when $A = S$. It follows immediately from Theorem 2.1.3 that:

Corollary 3.0.1. *If A is rational, and σ is coarser than (but including) the Nisnevich topology, then $\mathcal{H}^b(S) \subset \mathcal{H}^A_\sigma(S)$. Equivalently, L^A_σ equivalences are birational equivalences.*

3.1. Colimits of birational motives.

Theorem 3.1.1. *When σ is a topology coarser than the Nisnevich topology and A/S is a smooth rational scheme over S, the inclusion $\mathcal{H}^b(S) \subset \mathcal{H}^A_\sigma(S)$ is closed under sifted colimits. In particular, the inclusions $\mathcal{H}^b(S) \subset \mathcal{H}^{\mathbb{A}^1}(S)$ and $\mathcal{H}^b(S) \subset \mathcal{P}_\sigma(S)$ are closed under sifted colimits.*

Proof. Let us first consider the case of the discrete topology with $A = S$, in which case we are dealing with the inclusion $\mathcal{H}^b(S) \subset \mathcal{P}(S)$. Since the dense localization L_{dense} is defined by localizing morphisms of the site Sm_S, we have that $L_{dense}\mathcal{P}(S) \subset \mathcal{P}(S)$ is closed under all colimits. Indeed, dense localness is checked on sections, and colimits are computed sectionwise.

Since sifted colimits commute with finite products and presheaves in $\mathcal{P}_\Sigma(S)$ are defined in terms of taking coproducts to products, it is well known that $\mathcal{P}_\Sigma(S) \subset \mathcal{P}(S)$ is closed under sifted colimits. Since $\mathcal{H}^b(S) = \mathcal{P}_\Sigma(S) \bigcap L_{dense}\mathcal{P}(S)$, we are done in this case.

The case of a general topology σ, as in the statement, then follows formally. Indeed, consider a diagram $D : K \to \mathcal{H}^b(S)$ indexed by a sifted simplicial set K. Let $i : \mathcal{H}^b(S) \subset \mathcal{P}(S)$ be the inclusion and consider the colimit $P := \mathrm{colim}_K(i \circ D)$ in $\mathcal{P}(S)$. By the argument sketched above, P is birational local and hence $\mathrm{colim}_K D \simeq P$. To see that this is also the colimit in the σ local category, we compute the colimit in the σ local topos to be

$$\operatorname*{colim}_K i_\sigma \circ D = L_\sigma(\operatorname*{colim}_K iD) = L_\sigma P.$$

But since P is birational local, it is also Nisnevich local (see Theorem 2.1.3) and hence is σ local whenever σ is coarser than the Nisnevich topology. Therefore, the σ local colimit is identical (in $\mathcal{P}(S)$) to P itself.

The same trick now applies to $\mathcal{H}^A_\sigma(S)$, but with Corollary 3.0.1 instead of Theorem 2.1.3. □

Remark 3.1.2. A similar statement is certainly not true for the ordinary $\mathbb{A}^1$-motivic localization $\mathcal{H}^{\mathbb{A}^1}(S) \subset \mathcal{P}(S)$. Of course, $L_{\mathbb{A}^1}\mathcal{P}(S) \subset \mathcal{P}(S)$ is closed under all colimits. However, $\mathcal{P}_{nis}(S) \subset \mathcal{P}(S)$ is not closed under sifted colimits. This is primarily because the Nisnevich sheaf condition is given by taking Nisnevich cd squares to pullback squares, and sifted colimits do not commute with pullbacks. To make up for it, if we Nisnevich sheafify this, $\mathbb{A}^1$-locality is destroyed. Take, for example, $\mathbb{Z}(\mathbb{G}_m)$, the free abelian sheaf on Sm_k (where k is a perfect field) generated by $\mathbb{G}_m$

reduced at the unit section. $\mathrm{B}(\mathbb{Z}(\mathbb{G}_m))$ is not Nisnevich local but is $\mathbb{A}^1$-local. The Nisnevich sheafification of this last term is not $\mathbb{A}^1$-local (i.e., $\mathbb{Z}(\mathbb{G}_m)$ is not strongly $\mathbb{A}^1$-invariant, see [Cho14, Lemma 5.6]). What commutes with pullbacks, though, is filtered colimits. Thus, $\mathcal{H}^{\mathbb{A}^1}(S) \subset \mathcal{P}(S)$ is closed only under filtered colimits, not all sifted colimits.

Lemma 3.1.3. *Over a Qcqs base scheme S, let σ be a topology coarser than the Nisnevich topology yet finer than the disjoint union topology, and let A/S be a rational scheme. Then a model for the birational motivic localization functor $L_{bir} : \mathcal{P}(S) \to \mathcal{P}(S)$ is given by the formula: $L_{bir} \simeq \varinjlim_n (L^A_\sigma L_{dense})^n$. In particular,*

$$L_{bir} \simeq \varinjlim_n (L_\Sigma L_{dense})^n \simeq \varinjlim_n (L_{nis} L_{dense})^n \simeq \varinjlim_n (L_{mot} L_{dense})^n$$

Proof. This is standard. Let $\mathcal{X}$ be an S-space. Consider the colimit of the following diagram:

$$L_{dense}\mathcal{X} \to L^A_\sigma L_{dense}\mathcal{X} \to L_{dense}L^A_\sigma L_{dense}\mathcal{X} \to \cdots \tag{1}$$

The colimit in the statement is the colimit of the even-numbered subdiagram of the above diagram. Since the even-numbered subdiagram is cofinal in the original one, the first colimit in the statement is the same as the colimit of the above diagram 1. Now, the inclusion $\mathcal{P}_\Sigma(S) \subset \mathcal{P}(S)$ is closed under filtered colimits, and a σ sheaf is, in particular, a Σ-presheaf. Therefore, $\varinjlim_n (L^A_\sigma L_{dense})^n$ is a Σ-presheaf. But this colimit is also identical to the colimit of the odd-numbered subdiagram given by

$$L_{dense}\mathcal{X} \to L_{dense}L^A_\sigma L_{dense}\mathcal{X} \to \cdots \tag{2}$$

Since each term in this diagram is dense local by construction, so is the colimit. Thus, we obtain that $\varinjlim_n (L^A_\sigma L_{dense})^n\mathcal{X}$ is dense local. Combined with the previous paragraph, this implies that $\varinjlim_n (L^A_\sigma L_{dense})^n\mathcal{X}$ is birational local.

Because L^A_σ equivalences are, in particular, birational equivalences [Corollary 2.2.12], we see that each odd-numbered morphism in diagram 1, is a birational equivalence. On the other hand, the even-numbered morphisms are, by construction, dense equivalences and hence birational. It follows from the stability of the saturation class under colimits that the resulting morphism $\mathcal{X} \to \varinjlim_n (L^A_\sigma L_{dense})^n\mathcal{X}$ is a birational equivalence. □

Corollary 3.1.4. *$L_{bir} : \mathcal{P}(S) \to \mathcal{P}(S)$ preserves sifted colimits. More generally, for every topology σ coarser than the nisnevich topology, $L_{bir} : \mathcal{P}_\sigma(S) \to \mathcal{P}_\sigma(S)$ preserves sifted colimits.*

Proof. Recall that the functor L_Σ preserves sifted colimits [Lur09, Proposition 5.5.8.10(5)]. On the other hand, L_{dense} preserves all colimits. Indeed,

$$L_{dense} : \mathcal{P}(S) \to L_{dense}\mathcal{P}(S)$$

is a left adjoint and hence preserves all colimits, while $L_{dense}\mathcal{P}(S) \subset \mathcal{P}(S)$ is closed under all colimits. The first claim then follows from the formula

$$L_{bir} \simeq \varinjlim_n (L_\Sigma L_{dense})^n$$

and the fact that colimits commute with colimits. For a general topology σ, the general result then follows from the equivalences

$$L_{bir}L_\sigma \simeq L_{bir} \simeq L_\sigma L_{bir}.$$

□

In the next result, for a smooth scheme X/S, we denote by $h^b_S(X)$ the birational localization of the presheaf represented by $X/S \in Sm_S$, i.e., $h^b_S(X) := L_{bir}X = L_{bir}h_S X$.

Lemma 3.1.5. *Let S be Qcqs. For every $X \in Sm_S$, the birational space $h^b_S(X)$ is a compact projective object of $\mathcal{H}^b(S)$.*

Proof. Every smooth X/S is a compact projective object in $\mathcal{P}(Sm_S)$. Since projective objects are defined by preservation (by the corresponding representable functor) of geometric realizations, and compact objects by preservation of filtered colimits, we are done because $\mathcal{H}^b(S) \subset \mathcal{P}(S)$ is closed under these colimits by Theorem 3.1.1. □

Lemma 3.1.6. *Let S be a Qcqs scheme. Then, $\mathcal{H}^b(S)$ is a presentable ∞-category, generated under sifted colimits by the set*

$$\{h^b_S(X) |\ \textit{the structure map } p : X \to S \textit{ has property } P\},$$

where P is one of the following properties:

(a) When S is Qcqs: 1. smooth,
(b) When S is separated: 2. smooth affine, 3. smooth with affine domain;
(c) When S is the spectrum of a field of characteristic zero: 4. smooth projective.

Moreover, these colimits can actually be computed in $\mathcal{P}(S)$ itself.

Proof. By construction, $\mathcal{P}_\Sigma(Sm_S)$ is generated under sifted colimits by smooth X/S. Being its localization, so is $\mathcal{H}^b(S)$.

When S is separated, every $X \in Sm_S$ is separated by construction. So, by choosing Zariski covers (note that since S is Qcqs by Theorem 2.1.3, Zariski equivalences are birational), we may take the affines as generators from 1.

Finally, by Lemma 3.1.7 below, over a field of characteristic 0, every smooth variety admits a dense open immersion into a smooth proper variety. In this case, the 4th class is therefore dense local to the class in 1.

The last statement follows from Corollary 3.1.4. □

Lemma 3.1.7. *Let k be a field of characteristic zero. Then for every smooth X/k, there is a dense open immersion $X \to \tilde{X}$ into a smooth proper scheme $\tilde{X}/k$.*

Proof. By Nagata compactification, we can choose a dense open immersion into a proper variety $\bar{X}/k$. By Hironaka's theorem [Ann, Corollary 4.13], there exists a proper birational morphism $p : \widetilde{X} \to \overline{X}$ such that $\widetilde{X}$ is smooth and p is an isomorphism over the smooth locus of $\overline{X}$. Since X is smooth, the dense open $X \hookrightarrow \bar{X}$ lands in the smooth locus, and thus the pullback $p^{-1}(X) \to X$ is an isomorphism, yielding the desired dense open immersion $X \hookrightarrow \bar{X}$. □

Remark 3.1.8. Let us touch on a subtle point that one might naively expect to be true but fails because of how we have constructed our birational homotopy category. In §2 (and in the introduction), the birational homotopy category was suggested as a homotopical generalization of classical birational geometry. Thus, one would expect that all birational isomorphisms of schemes would be birational equivalences in this new world. The problem is that, since we started with Sm_k rather than Sch_k and are localizing only the dense open immersions belonging to Sm_k, this is not true by definition (and in fact, this fails in general as we shall show in Counterexample 4.2.14 in the 4th section). For example, since by assumption our $X/S \in Sm_S$ are separated and of finite type, and S is Qcqs, by Nagata compactification [Sta26, Theorem 0F41] we know that our X/S has compactifications. Thus, by taking the scheme-theoretic image in such a compactification, we obtain a dense $X \subset X'$ such that $X' \to S$ is proper. Assume that

this is equivalent (in $\mathcal{H}^b(S)$) to (the birational model of a) proper variety. Then the generating class property P as in the above theorem could be taken to be all proper schemes. Similarly, if S is Noetherian and X/S is proper, we use [Har77, Exercise 4.10] to find a projective X''/S with a birational morphism $X'' \to X$ over S. So, when S is Noetherian, one might expect that P can be taken to be projective. These, alas, are not clear over an arbitrary base.

3.2. **Preservation of finite limits and the bar construction.**

Proposition 3.2.1. *The inclusion $\mathcal{H}^b(S) \subset \mathcal{P}(S)$ is closed under all limits.*

Proof. This follows from the definition of local objects. □

The first key observation we make in this section is that the birational localization functor L_{bir} is cartesian.

Theorem 3.2.2. *Each possible composition of the functors*

$$\mathcal{P}(S) \xrightarrow{L_\Sigma} \mathcal{P}_\Sigma(S) \xrightarrow{L_{nis}} \mathcal{P}_{nis}(S) \xrightarrow{L_{mot}} \mathcal{H}^{\mathbb{A}^1}(S) \xrightarrow{L_{bir}} \mathcal{H}^b(S)$$

is cartesian.

Proof. Since L_Σ is the sheafification for the disjoint topology and L_{nis} is the sheafification for the Nisnevich topology, both are well known to be left exact. That L_{mot} is cartesian is a standard fact (see, for example, [Hoy17, Proposition 3.15]).

To conclude, it suffices to show that the composite $L_{bir} : \mathcal{P}(S) \to \mathcal{H}^b(S)$ is cartesian. Since filtered colimits commute with finite products and L_Σ preserves them, Lemma 3.1.3 immediately reduces the problem to showing that L_{dense} is cartesian.

Suppose X/S is smooth. The cartesian closedness of $\mathcal{P}(S)$ implies that the functor

$$h_S(X) \times - : \mathcal{P}(S) \to \mathcal{P}(S)$$

preserves colimits. Hence, to prove that it preserves $Dense(S)$, it suffices to show that it preserves dense open immersions. This is precisely Lemma 2.1.2. Again, since $\mathcal{P}(S)$ is generated under sifted colimits by smooth X/S and products commute with sifted colimits, we deduce that for any space $\mathcal{X} \in \mathcal{P}(S)$ the product functor

$$\mathcal{X} \times - : \mathcal{P}(S) \to \mathcal{P}(S)$$

preserves dense equivalences. Hence, if $\mathcal{Y}$ is another space in $\mathcal{P}(S)$, the composite

$$\mathcal{X} \times \mathcal{Y} \to \mathcal{X} \times L_{dense}\mathcal{Y} \to L_{dense}\mathcal{X} \times L_{dense}\mathcal{Y}$$

is a dense equivalence. But since $L_{dense}\mathcal{P}(S) \subset \mathcal{P}(S)$ is closed under limits, the above composite has a dense local space as its codomain. Hence, $L_{dense}\mathcal{X} \times L_{dense}\mathcal{Y}$ is a dense localization of $\mathcal{X} \times \mathcal{Y}$, meaning that

$$L_{dense}(\mathcal{X} \times \mathcal{Y}) \to L_{dense}\mathcal{X} \times L_{dense}\mathcal{Y}$$

is an equivalence. In other words, we have proved that L_{dense} preserves binary products.

(Since the terminal object $* \in \mathcal{P}(Sm_S)$ is already birational local, it is also the terminal object of $\mathcal{H}^b(S)$, and $L_{bir}(*) \simeq *$). □

Lemma 3.2.3. *L_{bir} has the following properties:*

(1) L_{bir} takes (group-like) (commutative) monoid S-spaces to (grouplike) (commutative) monoid S-spaces and thus restricts to

$$L_{bir} : (\mathrm{C})\mathrm{Mon}(\mathcal{P}(Sm_S))^{(gp)} \to (\mathrm{C})\mathrm{Mon}(\mathcal{H}^b(S))^{(gp)}$$

Similarly, L_{bir} preserves monoid actions.

(2) *When σ is coarser than the Nisnevich topology, L_{bir} commutes with the two sided σ local bar construction. In fact,*

$$L_{bir}\mathrm{B}_\sigma(-,-,-) \simeq \mathrm{B}(L_{bir}-, L_{bir}-, L_{bir}-).$$

In particular, if $\mathcal{M}$ is a birational monoid that acts on the birational local spaces $\mathcal{X}$ (from the right) and $\mathcal{Y}$ (from the left), then $\mathrm{B}_\sigma(\mathcal{X},\mathcal{M},\mathcal{Y})$ is birational local.

Proof. 1. Since L_{bir} is a cartesian localization, statement (1) is obvious.

2. First, assume that σ is the trivial topology. When $\mathcal{M}$ is a monoid acting on $\mathcal{X}$ (from the right) and on $\mathcal{Y}$ (from the left), the first statement implies that $L_{bir}\mathcal{M}$ is a monoid space acting on $L_{bir}\mathcal{X}$ and $L_{bir}\mathcal{Y}$, so $\mathrm{B}(L_{bir}\mathcal{X}, L_{bir}\mathcal{M}, L_{bir}\mathcal{Y})$ makes sense. Now, recall that

$$\mathrm{B}(\mathcal{X},\mathcal{M},\mathcal{Y}) := |\mathcal{X}\times\mathcal{M}^{\times\bullet}\times\mathcal{Y}|.$$

Since L_{bir} preserves finite products (see Theorem 3.2.2) and geometric realizations (Corollary 3.1.4), we have

$$\begin{aligned} L_{bir}(\mathrm{B}(\mathcal{X},\mathcal{M},\mathcal{Y})) &= L_{bir}|\mathcal{X}\times\mathcal{M}^{\times\bullet}\times\mathcal{Y}| \\ &\simeq \left|L_{bir}(\mathcal{X}\times\mathcal{M}^{\times\bullet}\times\mathcal{Y})\right| \text{ (since } L_{bir} \text{ preserves geometric realization)} \\ &\simeq \left|L_{bir}\mathcal{X}\times(L_{bir}\mathcal{M})^{\times\bullet}\times L_{bir}\mathcal{Y})\right| \text{ (since } L_{bir} \text{ preserves products)} \\ &= \mathrm{B}(L_{bir}\mathcal{X}, L_{bir}\mathcal{M}, L_{bir}\mathcal{Y}) \end{aligned}$$

When σ is a topology coarser than the Nisnevich topology, Theorem 2.1.3 implies that $L_{bir}L_\sigma \simeq L_{bir}$. Therefore,

$$L_{bir}\mathrm{B}_\sigma(-,-,-) \simeq L_{bir}L_\sigma\mathrm{B}(-,-,-) \simeq L_{bir}\mathrm{B}(-,-,-) \simeq \mathrm{B}(L_{bir}-, L_{bir}-, L_{bir}-).$$

For the first statement of (2), note that the object $\mathrm{B}(L_{bir}\mathcal{X}, L_{bir}\mathcal{M}, L_{bir}\mathcal{Y})$, being birational local already (as it is $\simeq L_{bir}(\mathrm{B}(\mathcal{X},\mathcal{M},\mathcal{Y}))$), is σ-local by Theorem 2.1.3 and hence

$$\mathrm{B}_\sigma(L_{bir}\mathcal{X}, L_{bir}\mathcal{M}, L_{bir}\mathcal{Y}) = L_\sigma\mathrm{B}(L_{bir}\mathcal{X}, L_{bir}\mathcal{M}, L_{bir}\mathcal{Y}) \simeq \mathrm{B}(L_{bir}\mathcal{X}, L_{bir}\mathcal{M}, L_{bir}\mathcal{Y}).$$

But in the previous paragraph, we have already identified this last term to $L_{bir}\mathrm{B}_\sigma(\mathcal{X},\mathcal{M},\mathcal{Y})$. □

Corollary 3.2.4. *When σ is coarser than the Nisnevich topology, L_{bir} commutes with the bar construction $\mathrm{B}_\sigma := L_\sigma\mathrm{B}(*,-,*)$, and thus the restriction functor*

$$L_{bir}\mathrm{B} : \mathrm{Mon}(\mathcal{H}^b(S)) \to \mathcal{H}^b(S)$$

coincides with

$$\mathrm{B}_{|\mathrm{Mon}(\mathcal{H}^b(S))}.$$

In particular, for a (commutative when $n \geq 2$) birational local monoid space $\mathcal{F}$, $\mathrm{B}^n_{nis}\mathcal{F}$ is again birational local.

Remark 3.2.5. This commutation relation is certainly false for L_{mot} and B^{nis}. One reason is that $\mathcal{H}^{mot} \subset \mathcal{P}^{nis}$ is not closed under geometric realizations; see Remark 3.1.2. It is worth noting that the problem has almost nothing to do with the $\mathbb{A}^1$-localization. In fact, it follows that $\mathrm{B}L_{\mathbb{A}^1} \simeq L_{\mathbb{A}^1}\mathrm{B}$. To see this, recall that $L_{\mathbb{A}^1}$ commutes with finite products [Hoy17, Corollary 3.5]. Moreover, being a localization at a schematic class, $L_{\mathbb{A}^1} : \mathcal{P}(S) \to \mathcal{P}(S)$ preserves all colimits (indeed, $L_{\mathbb{A}^1} : \mathcal{P}(S) \to L_{\mathbb{A}^1}\mathcal{P}(S)$ is a left adjoint, while $L_{\mathbb{A}^1}\mathcal{P}(S) \subset \mathcal{P}(S)$ is closed under all colimits). Since the bar construction is given by a geometric realization of products, and both commute with $L_{\mathbb{A}^1}$, we get $\mathrm{B}L_{\mathbb{A}^1} \simeq L_{\mathbb{A}^1}\mathrm{B}$ just as in Lemma 3.2.3. On the other hand, directed colimits also commute with the bar construction, so in the formulae $\varinjlim_n (L_{\mathbb{A}^1}\circ L_{nis})^n$ the only problematic candidate for this desired commutation is the nisnevich localization. Unfortunately, even though L_{nis} preserves products, it fails to preserve geometric realizations. In fact, it fails for any other topology finer than the $\sqcup$ topology.

Corollary 3.2.6. *The canonical functors*

$$\mathrm{K}(-,0)_{|Shv_S^b} := i_{disc} : Shv_S^b \to \mathcal{P}_\Sigma(Sm_S) \tag{3}$$

$$\mathrm{K}(-,1)_{|Grp_S^b} := \mathrm{B}_{|Grp_S^b} : Grp_S^b \to \mathcal{P}_\Sigma(Sm_S) \tag{4}$$

$$\mathrm{K}_{|Ab_S^b}(-,n) := \mathrm{B}^n_{|Ab_S^b} : Ab_S^b \to \mathcal{P}_\Sigma(Sm_S) \tag{5}$$

factors through the birational local subcategory $\mathcal{H}^b(S)$ and are identical to $\mathrm{K}^\sigma(-,n)$ for any σ coarser than the Nisnevich topology.

Corollary 3.2.7. *Birational (commutative) monoid spaces are (strictly) strongly $\mathbb{A}^1$-invariant. i.e.,*

$$\mathrm{Mon}(\mathcal{H}^b) \subset \mathrm{Mon}_{mot}(\mathcal{H}^{\mathbb{A}^1}) \text{ and } \mathrm{CMon}(\mathcal{H}^b) \subset \mathrm{CMon}_{mot}(\mathcal{H}^{\mathbb{A}^1}).$$

In particular,

$$Grp_S^b \subset Grp_S^{\mathbb{A}^1}, Ab_S^b \subset Ab_S^{\mathbb{A}^1},$$

where $Grp_S^{\mathbb{A}^1}$ (resp. $Ab_S^{\mathbb{A}^1}$) stand for strongly (resp. strictly) $\mathbb{A}^1$-invariant Nisnevich sheaves of groups (resp. abelian groups).

Proof. Since birational local spaces are $\mathbb{A}^1$-local, the result follows from Lemma 3.2.3. For example, let $\mathcal{M}$ be a birational monoid. By Corollary 3.2.4, $\mathrm{B}_{nis}\mathcal{M}$ is birational local. By Corollary 2.2.12, it follows that $\mathrm{B}_{nis}\mathcal{M}$ is motivic local. This is precisely the definition of strong $\mathbb{A}^1$ invariance of a monoid space [EHK$^+$21, Definition 3.1.6]. □

Remark 3.2.8. This strengthens Remark 2.3 in [BHS23]. In fact, our argument is entirely different from that in *loc.cit.* and works over all Qcqs base schemes.

Therefore, surprisingly, we see that defining strongly/strictly birational group sheaves (just like strongly/strictly $\mathbb{A}^1$-invariant group sheaves) is unnecessary.

Corollary 3.2.9. *For σ coarser than the Nisnevich topology, a σ-local presheaf of grouplike monoids $\mathcal{G}$ is birational if and only if $\mathrm{B}_\sigma\mathcal{G}$ is birationally local.*

Proof. The only-if part is just Corollary 3.2.4. For the if part, it suffices to observe that, because $\mathcal{G}$ is grouplike, $\mathcal{G} \simeq \Omega\mathrm{B}_\sigma\mathcal{G}$, and that Ω preserves local objects. □

Corollary 3.2.10. *The inclusion $Ab_S^b \subset Ab_S^\sigma$ is closed under all colimits and limits and, in particular, is exact. (See* [Mai26a, Corollary 3.2.3] *for a different argument.)*

Proof. Clearly, the inclusion is closed under all limits. Let us now show the right continuity of the inclusion. First, $L_{dense}Ab_S \subset Ab_S$ is closed under all colimits. By definition,

$$Ab_S^b = L_{dense}Ab_S \bigcap L_\Sigma Ab_S.$$

It thus suffices to show that $L_\Sigma Ab_S \subset Ab_S$ is also closed under all colimits. Since Ab_S is semiadditive, i.e., products are coproducts, this follows from the definition of L_Σ and the fact that taking the binary direct sum of abelian groups commutes with all colimits. □

Lemma 3.2.11. *When σ is coarser than the Nisnevich topology, L_{bir} preserves fiber sequences $\Gamma \to \mathcal{X} \to \mathcal{Y}$ of σ-local spaces with σ-locally connected base $\mathcal{Y}$.*

Proof. Every such fiber sequence is equivalent to a principal fiber sequence [AWW17, Theorem 2.3.3 Step 1, Step 2] by realizing $\mathcal{X} \simeq \mathbf{B}_\sigma(\Gamma, \Omega\mathcal{Y}, pt)$ and $\mathcal{Y} \simeq \mathbf{B}_\sigma\Omega\mathcal{Y}$.

$$\begin{array}{ccccc} \Gamma & \longrightarrow & \mathcal{X} & \longrightarrow & \mathcal{Y} \\ \downarrow\wr & & \downarrow\wr & & \downarrow\wr \\ \Gamma & \longrightarrow & \mathbf{B}_\sigma(\Gamma, \Omega\mathcal{Y}, pt) & \longrightarrow & \mathbf{B}_\sigma\Omega\mathcal{Y} \end{array}$$

Now we apply L_{bir} to the above diagram. Using Lemma 3.2.3 (2), the resulting diagram looks like:

$$\begin{array}{ccccc} L_{bir}\Gamma & \longrightarrow & L_{bir}\mathcal{X} & \longrightarrow & L_{bir}\mathcal{Y} \\ \downarrow\wr & & \downarrow\wr & & \downarrow\wr \\ L_{bir}\Gamma & \longrightarrow & \mathbf{B}_\sigma(L_{bir}\Gamma, L_{bir}\Omega\mathcal{Y}, pt) & \longrightarrow & \mathbf{B}_\sigma L_{bir}\Omega\mathcal{Y} \end{array}$$

But the bottom horizontal sequence of the diagram above is clearly a principal fiber sequence for the action of the group-like space $L_{bir}\Omega\mathcal{Y}$ on $L_{bir}\Gamma$.

It follows that the top horizontal sequence $L_{bir}\Gamma \to L_{bir}\mathcal{X} \to L_{bir}\mathcal{Y}$, being equivalent to the bottom one, is a fiber sequence as promised. □

Theorem 3.2.12. *When σ is coarser than the Nisnevich topology, the localization functor L_{bir} commutes with Ω on $\tau^\sigma_{\geq 1}\mathcal{P}_\sigma(Sm_S)_\bullet$.*

Proof. Apply the above lemma to the fiber sequence $\Omega\mathcal{E} \to pt \to \mathcal{E}$. □

Let us remind the reader that the motivic localization, even though not left exact, has a very nice property beyond cartesianness that makes certain motivic computations work well within $\mathcal{P}_{nis}(S)$ without taking motivic localizations. This is the property of local cartesianness of L_{mot}, which is the correct ∞-categorical analog of a right proper Bousfield localization. (Recall that a localization L of an ∞-category $\mathcal{C}$ is said to be locally cartesian if for every morphism $X \to Y$ of local objects and another morphism $Z \to X$ the canonical comparison map $L(Z \times_Y X) \to LZ \times_Y X$ is an equivalence in $\mathcal{C}$.)

Unfortunately, because dense open immersions are not stable under arbitrary pullbacks, L_{bir} is not locally cartesian:

Counterexample 3.2.13 (L_{bir} is not locally cartesian)**.** In this counterexample, we shall show that the birational localization functor is not locally cartesian. For this, let k be a field. Let X be an abelian k-variety of positive dimension. Note that X is birational local by Example 4.1.3. Let $0 : Speck \to X$ be the unit section and let $j : X \setminus 0 \hookrightarrow X$ be the inclusion at the complement at the origin. Clearly, j is dense. But the base change is the inclusion $\emptyset \hookrightarrow Speck$, which is evidently not a birational equivalence.

The following and its corollary are a weaker version of this property. We record them here for completeness and for a future use in [Mai26c, §4.4]. The proof below requires a birational connectivity result for L_{bir}, which is an immediate consequence of Lemma 3.2.3 and will be stated formally in Corollary 3.5.2.

Lemma 3.2.14. *L_{bir} is locally cartesian on σ-locally connected pointed objects. That is, if $\mathcal{X}$ is a σ-locally connected space, the morphism $\mathcal{Y} \to \mathcal{W}$ is in $\mathcal{H}^b(S)$, and $\mathcal{X} \to \mathcal{W}$ is given, then the morphism $L_{bir}(\mathcal{X} \times_{\mathcal{W}} \mathcal{Y}) \to L_{bir}(\mathcal{X}) \times_{\mathcal{W}} \mathcal{Y}$ is an equivalence.*

Proof. By factoring $\mathcal{X} \to \mathcal{W}$ through $L_{bir}\mathcal{X}$, we may assume that $\mathcal{X} \to \mathcal{W}$ is a birational weak equivalence, and thus $L_{bir}\mathcal{X} \simeq \mathcal{W}$. Let $\mathcal{Z} := \mathcal{X} \times_{\mathcal{W}} \mathcal{Y}$. We have to show that the canonical map

$$L_{bir}\mathcal{Z} \to L_{bir}\mathcal{X} \times_{L_{bir}\mathcal{X}} \mathcal{Y} \simeq \mathcal{Y}$$

is an equivalence. Let us consider the evident morphism of fiber sequences on the left and their birationalizations on the right:

$$\begin{array}{ccccc} \Gamma & \longrightarrow & \mathcal{Z} & \longrightarrow & \mathcal{X} \\ {\scriptstyle\phi}\downarrow & & \downarrow & & \downarrow \\ \Lambda & \longrightarrow & \mathcal{Y} & \longrightarrow & \mathcal{W} \end{array} \qquad \begin{array}{ccccc} L_{bir}\Gamma & \longrightarrow & L_{bir}\mathcal{Z} & \longrightarrow & L_{bir}\mathcal{X} \\ {\scriptstyle L_{bir}\phi}\downarrow & & \downarrow & & \downarrow \\ \Lambda & \longrightarrow & \mathcal{Y} & \longrightarrow & \mathcal{W} \end{array}$$

Since ϕ is an equivalence in $\mathcal{P}(S)$ (for example, by the pasting lemma for pullbacks) and Λ is birational local, we conclude that $L_{bir}(\Gamma) \simeq \Lambda$ (in fact, $\Gamma \simeq L_{bir}\Gamma$). Because L_{bir} preserves σ-locally connected fiber sequences [Lemma 3.2.11], the top line of the diagram on the right is a fiber sequence. The left-hand square in the same diagram (on the right) is a pullback square since the third leg $L_{bir}\mathcal{X} \to \mathcal{W}$ is an equivalence and the whole diagram is a morphism of fiber sequences. Moreover, the left bottom map $\Lambda \to \mathcal{Y}$ in that diagram is an effective epimorphism (since $\mathcal{W}$ is equivalent to $L_{bir}\mathcal{X}$, which is connected by birational connectivity Corollary 3.5.2). By [Lur09, Lemma 6.2.3.16], it is evident that the middle map $L_{bir}\mathcal{Z} \to \mathcal{Y}$ is an equivalence. This is exactly what we sought to prove. □

Corollary 3.2.15. *L_{bir} is locally cartesian on pointed $\mathbb{A}^1$-connected objects.*

Proof. Since L_{mot} is locally cartesian [Hoy17, Proposition 3.15] and $L_{bir}L_{mot} \simeq L_{bir}$ (see Corollary 2.2.12), we may apply L_{mot} to the whole setup and assume that $\mathcal{X}$ is Nisnevich connected. □

Remark 3.2.16. Since L_{bir} is not locally cartesian (Counterexample 3.2.13), it is in particular not left exact.

3.3. **Birationality vs Null motivic spaces.** In Lemma 3.1.3, we have constructed a standard formula for birational localization. However, without an explicit model for dense localization, the referred-to model is insufficient for meaningful computations. The aim of the next section is to show that when $\mathcal{X}$ is $\mathbb{A}^1$ connected and the base is a perfect field, there is an explicit model for $L_{bir}\mathcal{X}$, given by motivic $S^{1,1}$ and, equivalently, $S^{2,1}$ localization. (This idea of localizing the motivic homotopy category at motivic spheres has been studied extensively in [ABH23 §3] under the name 'null motivic spaces'.) To establish the connection between birational local spaces and $S^{2,1}$ (or $S^{1,1}$) null motivic spaces, we will need a few preliminary observations, which we present below.

Lemma 3.3.1. *Suppose $\Gamma \to \mathcal{X} \to \mathcal{Y}$ is a fiber sequence in $\mathcal{P}_\sigma(S)$, with $\mathcal{Y}$ a pointed σ(coarser than Nisnevich)-locally connected space. If Γ and $\mathcal{Y}$ are birational local, then so is $\mathcal{X}$.*

Proof. As in the proof of Lemma 3.2.11, we know that $\mathcal{X} \simeq \mathrm{B}_\sigma(\Gamma, \Omega\mathcal{Y}, pt)$, which is birational local by Lemma 3.2.3 (2), since both Γ and $\Omega\mathcal{Y}$ are birational local. □

Remark 3.3.2. When σ is non-trivial (for example, the Nisnevich topology), the standard method for proving such statements (see, for example, [Mor12, Lemma 6.51]) is not applicable here. Indeed, when the method in *loc.cit.* is adapted to the birational setup, it requires one to trace the following map of fiber sequences of Nisnevich-local spaces, induced by a non-trivial

dense open immersion $U \hookrightarrow X$:

$$\begin{array}{ccccc} \Gamma^X & \longrightarrow & \mathcal{X}^X & \longrightarrow & \mathcal{Y}^X \\ \downarrow{\scriptstyle j_\Gamma^*} & & \downarrow{\scriptstyle j_\mathcal{X}^*} & & \downarrow{\scriptstyle j_\mathcal{Y}^*} \\ \Gamma^U & \longrightarrow & \mathcal{X}^U & \longrightarrow & \mathcal{Y}^U \end{array}$$

But the problem is that even if $\mathcal{Y}$ is Nisnevich connected, there is no reason for either $\mathcal{Y}^X$ or $\mathcal{Y}^U$ to be so. So the morphism $\Gamma^U \to \mathcal{X}^U$ need not be an effective epimorphism.

Remark 3.3.3. There are non-birational presheaves of sets in $PSh_\Sigma(Sm_S)$ (and even in $PSh_\Sigma^{\mathbb{A}^1}(Sm_S)$) whose Nisnevich sheafification is birational. A general class of such examples is as follows: take a non-birational sheaf of groups G in $Grp_k^{\mathbb{A}^1}$ (say, $\mathbb{G}_m$). Let $F := H^1_{nis}(-; G)$ (for $\mathbb{G}_m$ this is the Picard group functor). Since G is strongly $\mathbb{A}^1$-invariant, we have $F(\mathbb{A}^1_X) = F(X)$, so $F \in PSh_\Sigma^{\mathbb{A}^1}(Sm_S)$. Moreover, its Nisnevich sheafification $L_{nis}F$ is trivial, hence birational. But clearly, F itself is not birational:

$$F(\mathbb{P}^1_k) = [\Sigma\mathbb{G}_m, \mathbf{B}^{nis}G]_{\mathcal{H}^{\mathbb{A}^1}} = [\mathbb{G}_m, G]_{\mathcal{H}^{\mathbb{A}^1}} = G(\mathbb{G}_m) = X^*G \neq pt = F(\mathbb{A}^1_k).$$

This kind of failure is also present in the case of $\mathbb{A}^1$-localization. Namely, there are non-$\mathbb{A}^1$-invariant presheaves of sets in $PSh_\Sigma(Sm_S)$ whose Nisnevich sheafification is $\mathbb{A}^1$-invariant. Here is an example. As above, take a non-$\mathbb{A}^1$-local Nisnevich sheaf G of groups in $PSh_\Sigma(Sm_S)$ (say, $\mathbb{G}_a$) over a proper scheme S. The presheaf $F := H^1_{zar}(-, \mathbb{G}_a)$ has trivial Nisnevich sheafification (since Zariski torsors are Nisnevich torsors as well), implying that F^{nis} is $\mathbb{A}^1$-local. But since $\mathbb{A}^1_\mathbb{Z}$ is affine, by Serre vanishing and the Kunneth formulae, one has

$$F(\mathbb{A}^1_S) = F(S) \otimes H^0_{zar}(\mathbb{A}^1_\mathbb{Z}, \mathbb{G}_a) = F(S)[t] \neq F(S).$$

Lemma 3.3.4. *Suppose S is a Qcqs scheme. For a space $\mathcal{X} \in \mathcal{P}(S)$, the following conditions are equivalent.*

(1) $\mathcal{X}$ is birational local.
(2) $\pi_i\mathcal{X}$ is birational local for all $i \geq 0$.
(3) $\tau_{\leq i}\mathcal{X}$ is birational local for all $i \geq 0$.

Proof. The equivalence 443(1) $\iff$ (2) is Proposition 2.1.8. Since $\pi_j\mathcal{X} = \pi_j\tau_{\leq i}\mathcal{X}$, for all $j \leq i$, the equivalence (3) $\iff$ (2) follows from the case of (1) $\iff$ (2). □

Proposition 3.3.5. *Suppose S is a Qcqs scheme and σ is a topology coarser than the Nisnevich topology. For a space $\mathcal{X} \in \mathcal{P}_\sigma(S)$, consider the following conditions:*

(1) $\mathcal{X}$ is birational local.
(2) $\tau^\sigma_{\leq i}\mathcal{X}$ is birational local for all $i \geq 0$.
(3) $\underline{\pi}^\sigma_i\mathcal{X}$ is a birational presheaf for all $i \geq 0$.

Then:

(a) $1 \implies 2 \implies 3$.
(b) If $\mathcal{X} \in \tau^\sigma_{\geq 1}\mathcal{P}_\sigma(Sm_S)_\bullet$ then $3 \implies 2$.
(c) If σ is postnikov complete then $2 \implies 1$.

In particular, when $\sigma = nis$, S is a Qcqs scheme of finite Krull dimension, and $\mathcal{X}$ is pointed and Nisnevich connected, the three conditions 1,2, and 3 are equivalent.

Proof. (a). Assume 1 holds. Then, by Lemma 3.3.4 we know that $\tau_{\leq i}\mathcal{X}$ is birational local. Since birational local spaces are automatically Nisnevich local [Theorem 2.1.3] it follows that $\tau_{\leq i}\mathcal{X}$ is Nisnevich local. And hence $\tau_{\leq i}\mathcal{X} \simeq L_\sigma\tau_{\leq i}\mathcal{X} =: \tau^\sigma_{\leq i}\mathcal{X}$. Sicne $\tau_{\leq i}\mathcal{X}$ is birational local, it follows that so is $\tau^\sigma_{\leq i}\mathcal{X}$. That is 2 holds.

Performing the same line of arguments with $\pi_i\mathcal{X}$ instead of $\tau_{\leq i}\mathcal{X}$ we see that $1 \implies 3$. We may now apply this $1 \implies 3$ result to the sheaves $\mathcal{Y}_i := \tau^\sigma_{\leq i}\mathcal{X}$ and obtain $2 \implies 3$.

(b). For the implications $3 \implies 2$ we use standard ideas of Postnikov tower. So assume 3 holds. First, consider the first slice of the Postnikov tower in the topos $\mathcal{P}_\sigma(S)$:

$$L_\sigma \mathrm{K}(\pi^\sigma_1(\mathcal{X}), 1) \to \tau^\sigma_{\leq 1}\mathcal{X} \to \tau^\sigma_{\leq 0}\mathcal{X} = pt$$

Note that, since $\pi^\sigma_1\mathcal{X}$ is given to be birational, by Corollary 3.2.6 we know that $\mathrm{K}^\sigma(\pi^\sigma_1(\mathcal{X}), 1)$ is birational local. Thus, the fiber in the above sequence is birational local. Since the base $\tau^\sigma_{\leq 0}\mathcal{X}$ is contractible (because $\mathcal{X}$ is σ-locally connecetd), it follows that $L_\sigma\mathbf{K}(\pi^\sigma_1(\mathcal{X}), 1) \simeq \tau^\sigma_{\leq 1}\mathcal{X}$, and therefore $\tau^\sigma_{\leq 1}\mathcal{X}$ is birational local. Inductively, assume that $\tau^\sigma_{\leq n}\mathcal{X}$ is birational local. We have the following fiber sequence:

$$\mathrm{K}^\sigma(\pi^\sigma_{n+1}(\mathcal{X}), n+1) \to \tau^\sigma_{\leq n+1}\mathcal{X} \to \tau^\sigma_{\leq n}\mathcal{X}$$

Since $\mathcal{X}$ is σ-locally connected, we know that so is $\tau^\sigma_{\leq n}\mathcal{X}$. On the other hand, since $\pi^\sigma_{n+1}\mathcal{X}$ is given to be birational local, by Corollary 3.2.6 we deduce that the fiber $\mathrm{K}^\sigma(\pi^\sigma_{n+1}(\mathcal{X}), n+1)$ is birational local. Thus, in the last fiber sequence, both the base and the fiber satisfy the conditions of Lemma 3.3.1, and we obtain the birational locality of $\tau^\sigma_{\leq n+1}\mathcal{X}$. By induction, we conclude that all the truncations $\tau^\sigma_{\leq n}\mathcal{X}$ are birational local. This proves $3 \implies 2$.

(c). It remains to prove that $3 \implies 1$ under the assumption that $\mathcal{P}_\sigma(S)$ is postnikov complete. To see how this works, assume all truncations $\tau^\sigma_{\leq n}\mathcal{X}$ are birational local. By Postnikov completeness of $\mathcal{P}_\sigma(S)$, we have $\mathcal{X} \simeq \varprojlim_n \tau^\sigma_{\leq n}\mathcal{X}$. Since the inclusion $\mathcal{H}^b(S) \subset \mathcal{P}_\sigma(S)$ is closed under limits [Proposition 3.2.1], it follows that

$$\varprojlim_n \tau^\sigma_{\leq n}\mathcal{X} \in \mathcal{H}^b(S),$$

and consequently, $\mathcal{X} \in \mathcal{H}^b(S)$.

For the 'in particular' part, use the well-known fact that when S is Qcqs of finite Krull dimension, the topos $\mathcal{P}_{nis}(S)$ is Postnikov complete. To see this, use [CM21, Theorem 3.18] and [Lur09, Proposition 7.2.1.10]. □

Remark 3.3.6. For the proof of (b), however, the methods of Remark 3.3.2 do work. The reason is that in this case $(\tau^{nis}_{\leq n}\mathcal{X})^X$ is connected. In fact, when $\tau^{nis}_{\leq n}\mathcal{X}$ is birational local, then so is $\pi^{nis}_n\tau^{nis}_{\leq n}\mathcal{X}$ (see Corollary 2.1.9). This implies, in particular, that this last term is birational local and hence flasque. Therefore, over a Noetherian base, it follows that

$$\pi_0(\tau^{nis}_{\leq n}\mathcal{X})^X \simeq H^n_{nis}(X \times -, \pi^{nis}_n\mathcal{X})$$

is trivial for all $n \geq 1$ (see [Rio02, Lemma 1.39 and 1.40]), so that $\pi^{nis}_0(\tau^{nis}_{\leq n}\mathcal{X})^X = *$.

Birationality and contraction

Let G be a presheaf of groups on a Qcqs scheme S. The $\mathbb{G}_m$-contraction of G is defined as

$$G_{-1} := \Omega_{\mathbb{G}_m^{\wedge 1}}G = ker\big(G(\mathbb{G}_{m,S} \times_S -) \xrightarrow{ev_1} G\big),$$

where the map $G(\mathbb{G}_{m,S} \times_S -) \xrightarrow{ev_1} G$ is the restriction along the unit section $1 : S \to \mathbb{G}_{m,S}$.

Lemma 3.3.7. *If G is a dense local presheaf of groups over S, then $G_{-1} = 0$.*

Proof. By Lemma 2.2.6, G is $\mathbb{G}_m$-local. That is $G(\mathbb{G}_{m,S} \times_S -) \to G$ is an isomorphism. Since G_{-1} is the kernel of this map, it is trivial. □

Remark 3.3.8. In the rest of the section, we will assume that the base is a perfect field. The reason is mostly generic smoothness. In particular, given a dense open immersion $U \hookrightarrow X$ one can find a filtration of X by open subschemes $U = U_n \subset U_{n-1} \subset \cdots \subset U_1 \subset U_0 = X$ such that for all i, $U_i \setminus U_{i-1} =: C_i$ is smooth and has trivial normal bundle in U_i (see [Remark 5.2, Lev10]). In fact, if $cod_X(X \setminus U) \geq d$, then it is clear that $cod_{U_i}(C_i) \geq d$ for all i. Thus, given a presheaf, any problem related to its injectivity or surjectivity of its restriction maps along open immersions can be reduced to checking the same on open immersions with smooth complement having a trivial normal bundle.

Lemma 3.3.9. *Suppose G is a presheaf of (not necessarily abelian) groups over a perfect field k. Then G is birational if and only if it is a strongly $\mathbb{A}^1$-invariant nisnevich sheaf with $G_{-1} = 0$.*

Proof. The only if part follows from Corollary 3.2.7 and Lemma 3.3.7.

Conversely, let G be a strongly $\mathbb{A}^1$-invariant Nisnevich sheaf of groups with $G_{-1} = 1$. Since such sheaves are weakly unramified [Bac24, Corollary 2.4], i.e., are injective on dense open immersions, to show that G is birational, it suffices to show that the restriction map $G(X) \to G(U)$ is surjective for a dense open immersion $U \hookrightarrow X$. Moreover, by Remark 3.3.8, we may assume that the complement $Z = X \setminus U$ is smooth and has a trivial normal bundle in X. In this situation, purity gives a cofiber sequence

$$Th_Z(N_Z X) \to \Sigma U \to \Sigma X,$$

where $N_Z X$ is the normal bundle of Z in X. Mapping this sequence to $\mathbf{B}^{nis}G$ (which is given to be $\mathbb{A}^1$-local) in $\mathcal{H}^{\mathbb{A}^1}(k)$, we are left to show that

$$[Th_Z(N_Z X), \mathbf{B}^{nis}G]_{\mathcal{H}^{\mathbb{A}^1}(k)}$$

is trivial. Since $U \hookrightarrow X$ is dense, the normal bundle has positive rank, say $n \geq 1$. Because Z has trivial normal bundle in X, i.e., there is an isomorphism $N_Z X \simeq Z \times \mathbb{A}^n_k$, we have

$$Th_Z(N_Z X) \simeq Z_+ \wedge T^n \simeq^{mot} \Sigma^n Z_+ \wedge \mathbb{G}^{\wedge n}.$$

So we see that indeed,

$$\begin{aligned}[Th_Z(N_Z X), \mathbf{B}^{nis}G]_{\mathcal{H}^{\mathbb{A}^1}(k)} &\simeq [\Sigma^n Z_+ \wedge \mathbb{G}^{\wedge n}, \mathbf{B}^{nis}G]_{\mathcal{H}^{\mathbb{A}^1}(k)} \simeq [\Sigma^{n-1} Z_+ \wedge \mathbb{G}^{\wedge n}, G]_{\mathcal{H}^{\mathbb{A}^1}(k)} \\ &\simeq [\Sigma^{n-1} Z_+ \wedge \mathbb{G}^{\wedge n-1}, \Omega_{\mathbb{G}} G]_{\mathcal{H}^{\mathbb{A}^1}(k)} \simeq pt \text{ (since } \Omega_{\mathbb{G}} G = G_{-1} = 1).\end{aligned}$$

□

Remark 3.3.10. If one has an independent proof of Corollary 4.6 in [BHS23], then, together with Lemma 3.3.9 here and Corollary 3.8 from *loc.cit.*, one may deduce Theorem 4.5 in *loc.cit.*

Motivic null spaces

We now recall the notion of motivic null spaces from [ABH23]. A motivic space is said to be $S^{p,q}$-null if it is local with respect to the motivic sphere $S^{p,q}$. In the special case of $S^{2,1}$ (and $S^{1,1}$), the definition can be restated as follows:

Definition 3.3.11. *A space $\mathcal{X} \in \mathcal{P}(S)$ is said to be motivically $S^{2,1}$ (resp. $S^{1,1}$) null if it is $\mathbb{A}^1$-local, Nisnevich-local, and $\mathbb{P}^1$-local (resp. $\mathbb{G}_m$-local) (see Definition 2.2.1). We denote the corresponding localization functors by*

$$L^{2,1} : \mathcal{P}(S) \to L^{2,1}\mathcal{P}(S) \text{ (and } L^{1,1} : \mathcal{P}(S) \to L^{1,1}\mathcal{P}(S), \text{ respectively).}$$

We now discuss the main result leading to the announced model of L_{bir}:

Theorem 3.3.12. *Let k be a perfect field. The following conditions on a Nisnevich-locally connected pointed motivic space $\mathcal{X}$ over k are equivalent:*

(1) $\mathcal{X}$ is birational local.
(2) $\tau^{nis}_{\leq i}\mathcal{X}$ is birational local for all $i \geq 0$.
(3) $\pi^{nis}_i\mathcal{X}$ is birational for all $i \geq 0$.
(4) $(\pi^{nis}_i\mathcal{X})_{-1} = 0$ for all $i \geq 0$.
(5) $\mathcal{X}$ is $\mathbb{G}_m$ local (equivalently, $S^{1,1}$ null).
(6) $\mathcal{X}$ is $\mathbb{P}^1$ local (equivalently, $S^{2,1}$ null).

Proof. The equivalence of the first three conditions is Proposition 3.3.5. (3) $\iff$ (4) follows from Lemma 3.3.9 since $\pi^{nis}_i\mathcal{X}$ are strongly $\mathbb{A}^1$-invariant [Mor12, Corollary 6.2]. (6) $\iff$ (4) $\iff$ (5) follows from [ABH23, Corollary 3.1.21]. Indeed, for a motivic space, $\mathbb{G}_m$-locality is equivalent to $S^{1,1}$-nullity, which, by the above-mentioned corollary [ABH23, Corollary 3.1.21], is equivalent to the vanishing $(\pi^{nis}_i\mathcal{X})_{-1} = 0$ for $i \geq 0$ (the case $i = 0$ is trivially true due to the connectivity of $\mathcal{X}$). Similarly, $\mathbb{P}^1$-locality is equivalent to $S^{2,1}$-nullity, which, by the conditions in *loc.cit*, is equivalent to the vanishing $(\pi^{nis}_i\mathcal{X})_{-1} = 0$ for $i \geq 1$ (again, the case $i = 0$ is trivially true by the connectivity of $\mathcal{X}$). □

Remark 3.3.13. Note that the results referred to in the proof, namely [ABH23, Corollary 3.1.21] do not use any explicit model for nullification and follow immediately from Morel's result: $\Omega_{\mathbb{G}}\pi^{\mathbb{A}^1}_i \simeq \pi^{\mathbb{A}^1}_i\Omega_{\mathbb{G}}$ [Mor12, Theorem 6.13].

Remark 3.3.14. Since

$$\mathcal{H}^b(k) \subset \mathcal{H}^{rat}(k) \subset \mathcal{H}^{\mathbb{A}^1}(k) \bigcap \mathcal{H}^{\mathbb{P}^1}(k)$$

(see Corollary 2.2.9), when k is a perfect field, the above theorem can be restated as saying that the inclusions restrict to equalities

$$\tau^{nis}_{\geq 1}\mathcal{H}^b(k) = \tau^{nis}_{\geq 1}\mathcal{H}^{rat}(k) = \tau^{nis}_{\geq 1}(\mathcal{H}^{\mathbb{A}^1}(k) \bigcap \mathcal{H}^{\mathbb{P}^1}(k)).$$

The following counterexample shows that these equalities need not hold without the connectivity assumption.

Counterexample 3.3.15 (to the possibility that $\mathcal{H}^{\mathbb{A}^1} \bigcap \mathcal{H}^{\mathbb{P}^1} \subset \mathcal{H}^b$)**.**

1. It is well known that the stack of finite etale schemes and isomorphisms, denoted Fét := FÉt$^{\simeq}$, is Nisnevich-local and $\mathbb{P}^1$-local (see, for example, [Mai26b, Example 2.2.18]). When the base is the field of complex numbers, we claim that this stack (viewed as a presheaf of groupoids on $Sm_{\mathbb{C}}$) is $\mathbb{A}^1$-local as well. To see this, it suffices to show that for any smooth scheme $X/\mathbb{C}$, the canonical map

$$\text{FÉt}(X) \to \text{FÉt}(X \times_{\mathbb{C}} \mathbb{A}^1_{\mathbb{C}})$$

induced by the projection $X \times_{\mathbb{C}} \mathbb{A}^1_{\mathbb{C}} \to X$ is an equivalence. This follows from the Riemann existence theorem [Mil80, Theorem 3.4] and the $\mathbb{C}$-invariance of finite coverings of topological spaces. However, Fét is not birational local. For example, consider the dense open immersion $\mathbb{G}_{m,\mathbb{C}} \hookrightarrow \mathbb{A}^1_{\mathbb{C}}$. By [Sta26, Lemma 0BQF], we know that FÉt$(\mathbb{A}^1_{\mathbb{C}}) \to$ FÉt$(\mathbb{G}_{m,\mathbb{C}})$ is fully faithful. Hence, if Fét$(\mathbb{A}^1_{\mathbb{C}}) \to$ Fét$(\mathbb{G}_{m,\mathbb{C}})$ were an equivalence of groupoids, then FÉt$(\mathbb{A}^1_{\mathbb{C}}) \to$ FÉt$(\mathbb{G}_{m,\mathbb{C}})$ would be so as well. But by the Riemann Existence Theorem again, this is impossible, since $\pi^{top}_1(\mathbb{G}_{m,\mathbb{C}}(\mathbb{C})) = \mathbb{Z}$ while $\pi^{top}_1(\mathbb{A}^1_{\mathbb{C}}(\mathbb{C})) = 0$. The problem is that Fét is not Nisnevich locally connected. Indeed,

$$\text{Fét}(\mathbb{C}) = \text{FÉt}(\mathbb{C})^{\simeq} = FinSet^{\simeq},$$

so π^{nis}_0FÉt evaluated at the base point $Spec\mathbb{C}$ equals to $\pi_0(FinSet^{\simeq}) = \mathbb{Z} \neq *$.

2. There is an even simpler counterexample. Namely, the scheme $\mathbb{G}_m$ is both $\mathbb{A}^1$-local and $\mathbb{P}^1$-local (see [Mai26b, Example 2.2.13]). But, $\mathbb{G}_m$ is not birational local; it is infact, birationally contractible. The prroblem is that $\mathbb{G}_m$ is disconnected.

Remark 3.3.16. The point of mentioning the first complecated example is the following:

1. First, $\mathbb{G}_m$ is discrete while Fét has higher homotopies.

2. Secondly, example (1) is also a cdh sheaf (infact, FÉt satisfies descent for proper surjecctive morphisms [Gro71, Theorem 4.12]. But $\mathbb{G}_m$ is not a cdh sheaf (for example, it is not a topological invariant). So the theorem is not true for non connected spaces even with cdh descent.

Remark 3.3.17. For grouplike spaces one does not need a connectivity assumption. We shall establish this in [Mai26b].

Remark 3.3.18. It is worth noting that, stably, a similar claim holds without any connectivity assumption (i.e., an S^1-spectra is birational local iff it is $\mathbb{A}^1$-local and $\mathbb{P}^1$-local). The version for chain comlexes with transfers is known due to Ayoub [Ayo20, Proposition 3.9].

3.4. **On finding a model for L_{bir}.** In this section, we will finally construct a model for L_{bir} on motivically connected spaces. Before that, we take a detour to establish the required connectivity result for null spaces. The authors of [ABH23] deduce a connectivity result for $L^{p,q}$ in general; however, their formula for $L^{p,q}$, denoted $(-)^{p,q}_{\infty}$, is obtained only via the small object argument. We emphasize that in the special case of $S^{2,1}$ (and $S^{1,1}$) localization, it is possible to construct an explicit model (over an arbitrary base), and, in fact, one may immediately deduce a connectivity result from it. In an upcoming work, we will study the $\mathbb{P}^1$ motivic homotopy theory in greater detail [PHT Mai26b]. We recall the relevant results from there and note that they apply equally to $\mathbb{G}_m$ localization and, for that matter, to localization at any smooth scheme with a rational point. The reader may use the connectivity result in [ABH23, Corollary 3.1.26]. Our result is stronger in that it holds over arbitrary base schemes and yields an explicit surjection of path components.

Let us recall the following results from [Mai26b]:

Theorem 3.4.1 (Mai26b, Theorem 2.1.1, Lemma 2.1.4)**.** *Let Y/S be a smooth scheme and $\mathcal{X} \in \mathcal{P}(S)$.*

(1) Let $\delta^{\bullet}_{\mathbb{P}^1} \times Y$ be the sub-diagram of the slice category $(Sm_S)_{/Y}$ defined by

$$\delta^{n}_{\mathbb{P}^1} \times Y := (\mathbb{P}^1_S)^{\times n} \times_S Y,$$

with structure maps given by projection to Y. The $\mathbb{P}^1$-localization functor on $\mathcal{P}(S)$ is given by the following colimit formulae:

$$X \mapsto L_{\mathbb{P}^1}\mathcal{X}(Y) = \operatorname*{colim}_{(\delta^{\bullet}_{\mathbb{P}^1} \times Y)^{op}} \mathcal{X}(\delta^{n}_{\mathbb{P}^1} \times Y)$$

(2) The canonical map $\pi_0\mathcal{X} \to \pi_0 L_{\mathbb{P}^1}\mathcal{X}$ is a surjective morphism of presheaves of sets on Sm_S.

This leads to the following explicit model of $L^{2,1}$:

Theorem 3.4.2. *A model for $S^{2,1}$ motivic localization is given by :*

$$L^{2,1} := L^{\mathbb{A}^1,\mathbb{P}^1}_{nis} \simeq \operatorname*{colim}_{n\to\infty}(L_{nis} \circ L_{\mathbb{A}^1} \circ L_{\mathbb{P}^1})^n \simeq \operatorname*{colim}_{n\to\infty}(L^{\mathbb{A}^1}_{nis} \circ L^{\mathbb{P}^1}_{nis})^n \simeq \operatorname*{colim}_{n\to\infty}(L^{\mathbb{P}^1}_{nis} \circ L^{\mathbb{A}^1}_{nis})^n$$

where $L_{\mathbb{P}^1}\mathcal{X}(Y) = \operatorname{colim}_{(\delta^{\bullet}_{\mathbb{P}^1} \times Y)^{op}} \mathcal{X}(\delta^{m}_{\mathbb{P}^1_S} \times_S Y)$ and $L_{\mathbb{A}^1}\mathcal{Y}(Y) = |\mathcal{Y}(\Delta^{\bullet}_{\mathbb{A}^1} \times Y)|$.

Proof. The method is similar to Lemma 3.1.3. The space $\operatorname{colim}_{n\to\infty}(L^{\mathbb{A}^1}_{nis}\circ L^{\mathbb{P}^1}_{nis})^n\mathcal{X}$, associated with an S-space $\mathcal{X}$, is the colimit of the even-numbered subdiagram of the following diagram:

$$L^{\mathbb{P}^1}_{nis}\mathcal{X}\to L^{\mathbb{A}^1}_{nis}L^{\mathbb{P}^1}_{nis}\mathcal{X}\to L^{\mathbb{P}^1}_{nis}L^{\mathbb{A}^1}_{nis}L^{\mathbb{P}^1}_{nis}\mathcal{X}\to\cdots$$

It follows that the colimit of the whole diagram is both $\mathbb{A}^1$-local and Nisnevich-local. Similarly, by considering the odd-numbered (cofinal) subdiagram, it follows that the colimit is $\mathbb{P}^1$-local as well. The same arguments apply verbatim to the other colimit $\operatorname{colim}_{n\to\infty}(L^{\mathbb{P}^1}_{nis}\circ L^{\mathbb{A}^1}_{nis})^n\mathcal{X}$.

For the first colimit, one considers the $3\mathbb{N}$, $3\mathbb{N}+1$, and $3\mathbb{N}+2$ subdiagrams of the diagram:

$$L_{\mathbb{P}^1}\mathcal{X}\to L_{\mathbb{A}^1}L_{\mathbb{P}^1}\mathcal{X}\to L_{nis}L_{\mathbb{A}^1}L_{\mathbb{P}^1}\mathcal{X}\to L_{\mathbb{P}^1}L_{nis}L_{\mathbb{A}^1}L_{\mathbb{P}^1}\mathcal{X}\to L_{\mathbb{A}^1}L_{\mathbb{P}^1}L_{nis}L_{\mathbb{A}^1}L_B\mathcal{X}\to\cdots$$

and proceeds similarly. □

Corollary 3.4.3 (see [ABH23, Corollary 3.1.26] for a weaker version)**.** *Let S be Qcqs and $\mathcal{X}\in\mathcal{P}(S)$. Then there is a canonical surjection $\pi_0^{nis}\mathcal{X}\to\pi_0^{nis}L^{2,1}\mathcal{X}$. This implies that $L^{2,1}$ preserves Nisnevich connectivity.*

Proof. From Theorem 3.4.1 (2), it follows that $\pi_0^{nis}\mathcal{X}\to\pi_0^{nis}L_{\mathbb{P}^1}\mathcal{X}$ is surjective (see [Mai26b, Proposition 2.1.5] for details). By [MV99, Corollary 2.3.22], $\pi_0^{nis}\mathcal{X}\to\pi_0^{nis}L_{\mathbb{A}^1}\mathcal{X}$ is also surjective.

The claim then follows immediately from the formulas for $L^{2,1}$ given in the previous theorem, Theorem 3.4.2. For example, $L^{2,1}\mathcal{X}$ is given by the colimit of the following diagram:

$$\mathcal{X}\to L^{\mathbb{P}^1}_{nis}\mathcal{X}\to L^{\mathbb{A}^1}_{nis}L^{\mathbb{P}^1}_{nis}\mathcal{X}\to L^{\mathbb{P}^1}_{nis}L^{\mathbb{A}^1}_{nis}L^{\mathbb{P}^1}_{nis}\mathcal{X}\to\cdots$$

Since π_0^{nis} preserves colimits, we see that $\pi_0^{nis}L^{2,1}\mathcal{X}$ is the colimit of the following diagram

$$\pi_0^{nis}\mathcal{X}\to\pi_0^{nis}L^{\mathbb{P}^1}_{nis}\mathcal{X}\to\pi_0^{nis}L^{\mathbb{A}^1}_{nis}L^{\mathbb{P}^1}_{nis}\mathcal{X}\to\pi_0^{nis}L^{\mathbb{P}^1}_{nis}L^{\mathbb{A}^1}_{nis}L^{\mathbb{P}^1}_{nis}\mathcal{X}\to\cdots$$

Now, inductively applying the observations from the previous paragraph, we know that each map in the last diagram is an epimorphism of Nisnevich sheaves of sets. We conclude that the colimit $\pi_0^{nis}\mathcal{X}\to\pi_0^{nis}L^{2,1}\mathcal{X}$ is also an epimorphism. □

Theorem 3.4.4. *Over a perfect field k, the restriction of*

$$L_{bir}:\mathcal{P}(k)_\bullet\to\mathcal{H}^b(k)$$

to $L^{2,1}$-connected objects (resp. $L^{1,1}$-connected objects) is equivalent to the restriction of $L^{2,1}$ (resp. $L^{1,1}$) to those objects.

Proof. Suppose $\mathcal{X}$ is a space with a chosen base point such that $L^{2,1}\mathcal{X}$ (or $L^{1,1}\mathcal{X}$) is Nisnevich connected. By Theorem 3.3.12, $L^{2,1}\mathcal{X}$ (or $L^{1,1}\mathcal{X}$) is birational local. To conclude, it suffices to note that $L^{2,1}$ (or $L^{1,1}$) equivalences are birational equivalences. Indeed, then the localization map $\mathcal{X}\to L^{2,1}\mathcal{X}$ (or $\mathcal{X}\to L^{1,1}\mathcal{X}$) is a birational equivalence to a birational local space, and hence is a model of birational localization. But since $\mathbb{A}^1,\mathbb{P}^1,\mathbb{G}_m$ are rational it is obvious that $L^{2,1}$ and $L^{1,1}$ equivalences are birational equivalences [Corollary 2.2.9 (1)]. □

Corollary 3.4.5. *In the previous theorem, it suffices to assume that $\mathcal{X}$ is $\mathbb{A}^1$-connected, $\mathbb{P}^1$-connected, or even Nisnevich-connected.*

Proof. By Corollary 3.4.3, Nisnevich connectivity implies $L^{2,1}$-connectivity. When $\mathcal{X}$ is $\mathbb{A}^1$-connected, replace it with the Nisnevich-connected space $L_{mot}\mathcal{X}$. Similarly, when $\mathcal{X}$ is $\mathbb{P}^1$-connected, replace it with the Nisnevich-connected space $L_{pmot}\mathcal{X}$. □

Corollary 3.4.6. *On pointed $\mathbb{A}^1$-connected spaces over perfect fields, the birational localization functor coincides with $L^{2,1}$ and $L^{1,1}$. Thus,*

$$L_{bir}\big|_{\tau^{nis}_{\geq 1}\mathcal{H}^{\mathbb{A}^1}(k)} \simeq \operatorname*{colim}_{n\to\infty}(L_{nis}\circ L_{\mathbb{A}^1}\circ L_{\mathbb{P}^1})^n\big|_{\tau^{nis}_{\geq 1}\mathcal{H}^{\mathbb{A}^1}} \simeq \operatorname*{colim}_{n\to\infty}(L_{nis}\circ L_{\mathbb{A}^1}\circ L_{\mathbb{G}_m})^n\big|_{\tau^{nis}_{\geq 1}\mathcal{H}^{\mathbb{A}^1}}$$

where

$$L_{\mathbb{P}^1}\mathcal{X}(Y) = \operatorname*{colim}_{(\delta^n_{\mathbb{P}^1}\times Y)^{op}} \mathcal{X}(\delta^m_{\mathbb{P}^1}\times Y) \text{ and } L_{\mathbb{A}^1}\mathcal{Y}(Y) = |\mathcal{Y}(\Delta^\bullet_{\mathbb{A}^1}\times Y)|.$$

It follows that L_{bir} takes $\mathbb{A}^1$ connected spaces to Nisnevich connected spaces.

Corollary 3.4.7. *Over a perfect field, the birational localization functor preserves Nisnevich local connectivity.*

Proof. If $\mathcal{X}$ is Nisnevich connected, by Corollary 3.4.5, a choice of a base point of $\mathcal{X}$ (see Remark 3.5.8) yields $L_{bir}\mathcal{X} \simeq L^{2,1}\mathcal{X}$. But Corollary 3.4.3 implies that $L^{2,1}\mathcal{X}$ is Nisnevich connected. Thus $L_{bir}\mathcal{X}$ is Nisnevich connected as well. □

Remark 3.4.8. From Construction 3.1.17 of [ABH23], we do have a formula for $S^{2,1}$ or $S^{1,1}$ localization, and hence,

$$L_{bir}\big|_{\tau^{nis}_{\geq 1}\mathcal{H}^{\mathbb{A}^1}} \simeq [(-)^{1,1}_\infty]\big|_{\tau^{nis}_{\geq 1}\mathcal{H}^{\mathbb{A}^1}} \simeq [(-)^{2,1}_\infty]\big|_{\tau^{nis}_{\geq 1}\mathcal{H}^{\mathbb{A}^1}}$$

Note, however, that these formulas are not explicit and are obtained using a small-object argument.

3.5. **Connectivity of L_{bir}.** In Corollary 3.4.7 of the last section, we saw that over a perfect field, L_{bir} preserves Nisnevich connectivity. Let us now compare this to the case of L_{mot}. Recall that, over an arbitrary *Qcqs* base, the singular model of motivic localization yields a canonical surjection $\pi_0^{nis} \to \pi_0^{\mathbb{A}^1}$, which implies a connectivity property for L_{mot} over an arbitrary Qcqs base scheme. For birational localization, we constructed a model in the previous section. However, it has two primary drawbacks: first, it works only for connected spaces, and second, it works only over perfect fields. Unfortunately, it seems less plausible to have such a mechanism for all spaces or over all bases. In this section, we will take an entirely different route to prove, in fact, a stronger version of connectivity for the birational localization functor. The key will be the commutation of L_{bir} with the bar construction, as obtained in §3.2. We continue to assume that S is Qcqs. We start with the primary observation,

Corollary 3.5.1. *A birational local space is birationally connected if and only if it is $\mathbb{A}^1$-connected if and only if it is Nisnevich-connected if and only if it is sectionwise connected.*

Proof. Clearly, if $\mathcal{X}$ is birational local, then

$$\pi_0^b\mathcal{X} \cong \pi_0\mathcal{X} \cong \pi_0^{nis}\mathcal{X}$$

(see Corollary 2.1.9 for the last isomorphism). By Corollary 2.2.12, it follows that $\mathcal{X} \simeq L_{mot}\mathcal{X}$. Applying π_0^{nis} to this equivalence yields $\pi_0^{nis}\mathcal{X} \cong \pi_0^{\mathbb{A}^1}\mathcal{X}$. Thus

$$\pi_0^b\mathcal{X} \cong \pi_0\mathcal{X} \cong \pi_0^{nis}\mathcal{X} \cong \pi_0^{\mathbb{A}^1}\mathcal{X}.$$

□

Thus, we shall say that a space $\mathcal{X}$ is birationally connected if $L_{bir}\mathcal{X}$ is (sectionwise/Nisnevich locally/$\mathbb{A}^1$)-connected.

Corollary 3.5.2. *If σ is a topology coarser than the Nisnevich topology over an arbitrary Qcqs base S, then L_{bir} takes σ locally $n(\geq 0)$-connected pointed S-spaces to sectionwise n-connected S-spaces.*

Proof. From the loop-path fiber sequence, the existence of a base point for $\mathcal{X} \in \tau_{\geq 1}\mathcal{P}(S)$ yields an equivalence $\mathcal{X} \simeq \mathrm{B}_\sigma\Omega(\mathcal{X})$. By induction, if $\mathcal{X}$ is σ-locally n-connected, then $\mathcal{X} \simeq \mathrm{B}^{n+1}_\sigma\Omega^{n+1}\mathcal{X}$. Hence,

$$L_{bir}\mathcal{X} \simeq L_{bir}\mathrm{B}^{n+1}_\sigma\Omega(\mathcal{X}) \simeq \mathrm{B}^{n+1}L_{bir}\Omega(\mathcal{X})$$

[by Lemma 3.2.3]. By construction, B^{n+1} takes values in (sectionwise) n-connected objects and we are done. □

Corollary 3.5.3. *L_{bir} takes pointed $\mathbb{A}^1$-n-connected spaces to (sectionwise) n-connected spaces. In particular, pointed $\mathbb{A}^1$-connected spaces are birationally connected.*

Proof. Assume $\mathcal{X}$ is pointed $\mathbb{A}^1$-n-connected, i.e., $\mathcal{Y} := L_{mot}\mathcal{X}$ is a Nisnevich n-connected pointed space. By the previous corollary applied to the Nisnevich topology, $L_{bir}\mathcal{Y}$ is sectionwise n-connected. Since motivic equivalences are birational equivalences (see Corollary 2.2.12), we have

$$L_{bir}\mathcal{X} \simeq L_{bir}L_{mot}\mathcal{X} = L_{bir}\mathcal{Y}.$$

It thus follows that $L_{bir}\mathcal{X}$ is sectionwise n-connected. □

Remark 3.5.4 ((-1, -2)-connectivity)**.** Since every object is (-2)-connected, like any localization, L_{bir} preserves (-2)-connectivity. Similarly, a (-1)-connectivity statement is also true for any localization. For example, if $U \to \mathcal{X}$ is a non trivial section of a space $\mathcal{X}$ over some nonempty scheme U, then $U \to \mathcal{X} \to L_{bir}\mathcal{X}$ provides a (non-empty) section of $L_{bir}\mathcal{X}$.

Remark 3.5.5. One reason such a method of proving connectivity, as in Corollary 3.5.2, is not feasible in the motivic case is that the 2nd statement in Lemma 3.2.3 fails in the $\mathbb{A}^1$-motivic case, see Remark 3.2.5. This is why, in the $\mathbb{A}^1$-motivic case, an explicit formula for the motivic localization functor, which ensures surjectivity on π_0^{nis}, was needed to obtain an unstable connectivity result [MV99, Corollary 3.22].

Remark 3.5.6. The surjectivity of π_0^{nis} for the $\mathbb{A}^1$-motivic localization [MV99, Corollary 3.22] ensures that $\eta : 1 \to L_{mot}$ is an objectwise effective epimorphism, an important tool for transferring several results from the Nisnevich topos to the motivic homotopy category (see, for example, [Mai26c, Corollary 3.4.5]). However, for birational localization, the canonical map

$$\pi_0^{nis}\mathcal{X} \to \pi_0^{nis}L_{bir}\mathcal{X} = \pi_0 L_{bir}\mathcal{X}$$

is not surjective (see the counterexample below), unlike its $\mathbb{A}^1$-motivic counterpart. As a result, the birational localization yields weaker results (see [Mai26c, Corollary 4.4.6]).

Counterexample 3.5.7. In this counterexample, we show that L_{bir} is not an effective localization (of the Nisnevich ∞-topos), i.e.,

$$\pi_0^{nis}\mathcal{X} \to \pi_0^{nis}L_{bir}\mathcal{X}$$

need not be an epimorphism of sheaves in general. Of course, over finite fields, it is easy to find counterexamples:

(1) Let $S = Speck$ for $k = \mathbb{F}_q$. Consider the rational scheme $X = \mathbb{P}^1_{\mathbb{F}_q} \setminus \mathbb{P}(\mathbb{F}_q)$. Since $\mathbb{P}(\mathbb{F}_q)$ is a finite set of points, X is a dense open subscheme of $\mathbb{P}^1_{\mathbb{F}_q}$ and hence birationally contractible. Thus, the map in question can be identified with the canonical projection $X \to S$, which is clearly not surjective, since, by construction, X has no k-rational points. This example certainly fails for infinite fields.
(2) For an example that works in all scenarios, consider the Counterexample 3.2.13. That is, let X be an abelian variety over k of positive dimension, and let $U := X \setminus 0$ be the dense open subscheme of X given by the complement of the unit. Since X is birationally rigid (see

Example 4.1.3), it is clear that $L_{bir}U \simeq X$. So the morphism of path component sheaves of the birational localization of U is identified with the map

$$\pi_0^{nis}U = U \hookrightarrow X = \pi_0^{nis}X \simeq \pi_0^{nis}L_{bir}U,$$

which not only fails to be surjective but is in fact an injection.

Remark 3.5.8. When the topology σ is such that the base scheme S admits no nontrivial σ-cover, we can remove the pointed condition from each of the statements above. This is always the case when S is a point for the topology σ. For example, this holds for every S when σ is the discrete topology, when $\sigma = zar$ and S is the spectrum of a local ring, or when $\sigma = nis$ and S is the spectrum of a henselian local ring. Indeed, in these cases $\pi_0^\sigma\mathcal{X}(S) = \pi_0\mathcal{X}(S)$. However, this is not always the case. For example, this might fail when $S = \mathbb{P}^1_k$ and $\sigma = zar$. An example of a topology finer than the Nisnevich topology where this fails is the étale topology over the spectrum of a non-separably closed field.

Corollary 3.5.9. *Let S be the spectrum of a henselian local ring. Then,*

(1) L_{bir} takes Nisnevich locally n-connected S-spaces to sectionwise n-connected S-spaces.
(2) L_{bir} takes $\mathbb{A}^1$-n-connected S-spaces to sectionwise n-connected S-spaces.
(3) $\mathbb{A}^1$-connected spaces over S are birationally connected.

Proof. Because S is henselian, for any $\mathcal{X} \in \mathcal{P}(S)$ we have $\pi_0\mathcal{X}(S) \simeq \pi_0^{nis}\mathcal{X}(S)$. Hence, if $\mathcal{X}$ is Nisnevich connected, then

$$\pi_0\mathcal{X}(S) \simeq \pi_0^{nis}\mathcal{X}(S) = *.$$

We may thus choose a section $S \to \mathcal{X}$ to make $\mathcal{X}$ pointed. So (1) follows from Corollary 3.5.2. For (2), use $\mathcal{Y} := L_{mot}\mathcal{X}$ as in the proof of Corollary 3.5.2. (3) is a special case of (2). □

3.6. **Birational homotopy groups of spaces.** We will use the notation $\pi_i^b := \pi_i L_{bir}$ and call these the birational homotopy group presheaves. By Proposition 2.1.8, these presheaves on Sm_S (see Definition Definition 2.1.6) are therefore automatically Nisnevich sheaves by Theorem 2.1.3. Consequently, we have $\pi_i^b = \pi_i^{nis}L_{bir}$, aligning with the conventions in motivic homotopy theory. Throughout this section, we assume that σ is a topology coarser than the Nisnevich topology.

Proposition 3.6.1. *$\pi_0^b : \mathcal{P}(S) \to PSh_S$ preserves sifted colimits.*

Proof. Since π_0 preserves colimits this follows from the fact that L_{bir} preserves sifted colimits (Theorem 3.1.1). □

Birational connective covers and universal properties of π_i^b:

Let $\mathcal{X} \in \mathcal{P}_\sigma(S)_\bullet$ and

$$\tau^\sigma_{>n}\mathcal{X} \to \mathcal{X} \to \tau^\sigma_{\leq n}\mathcal{X}$$

be the n-th slice of the Postnikov tower of $\mathcal{X}$.

Lemma 3.6.2. *If $\mathcal{X}$ is a birational local space, then $\tau^\sigma_{>n}\mathcal{X}$ is birational local.*

Proof. When $\mathcal{X}$ is birational local, it follows that so is $\tau_{>1}\mathcal{X}$ (use, for example, Proposition 2.1.8). But then, for any topology σ coarser than the nisnevich topology, $\tau_{>1}\mathcal{X}$ is σ local as well [Theorem 2.1.3]. Therefore,

$$\tau^\sigma_{>n}\mathcal{X} := L_\sigma\tau_{>n}\mathcal{X} \simeq \tau_{>1}\mathcal{X},$$

and this last term is already birational local. □

Corollary 3.6.3. *The (sectionwise) simply connected covering space of a birational local space is birational local. We may call this covering space a 'birationally simply connected cover'.*

Proof. This is the $n = 1$ case of the lemma above. □

Since $\tau^\sigma_{>n}L_{bir}$ is birational local by Lemma 3.6.2, the natural map $\tau^\sigma_{>n} \to \tau^\sigma_{>n}L_{bir}$ descends to a map

$$L_{bir}\tau^\sigma_{>n} \to \tau^\sigma_{>n}L_{bir}.$$

Similarly, by Proposition 3.3.5 there is a canonical map $L_{bir}\tau^\sigma_{\leq n} \to \tau^\sigma_{\leq n}L_{bir}$.

Proposition 3.6.4. *Let $\mathcal{X}$ be a pointed σ-locally 0-connected S-space such that for some $n \geq 0$ the S-space $\tau^\sigma_{\leq n}\mathcal{X}$ is birational (this happens precisely when $\pi^\sigma_i\mathcal{X}$ are birational local for all $i \leq n$, see Corollary 2.1.9). Then, for all $i \leq n$, the canonical morphisms*

$$L_{bir}\tau^\sigma_{\leq i} \to \tau^\sigma_{\leq i}L_{bir} \text{ and } L_{bir}\tau^\sigma_{>i} \to \tau^\sigma_{>i}L_{bir}$$

are equivalences.

Proof. Since $\tau^\sigma_{\leq i}\mathcal{X}$ is σ-locally connected, we obtain a fiber sequence

$$L_{bir}\tau^\sigma_{>i}\mathcal{X} \to L_{bir}\mathcal{X} \to L_{bir}\tau^\sigma_{\leq i}\mathcal{X}$$

by applying Lemma 3.2.11. But $L_{bir}\tau^\sigma_{\leq i}\mathcal{X} \simeq \tau^\sigma_{\leq i}\mathcal{X}$ for $i \leq n$ because $\tau^\sigma_{\leq n}\mathcal{X}$ is birational. Because $\tau^\sigma_{\leq i}\mathcal{X}$ is connected and $\tau^\sigma_{>i}\mathcal{X}$ is i-connected (in $\mathcal{P}_\sigma(S)$), by higher connectivity [Corollary 3.5.2], $L_{bir}\tau^\sigma_{\leq i}\mathcal{X}$ is connected and $L_{bir}\tau^\sigma_{>i}\mathcal{X}$ is i-connected. From the long exact sequence of homotopy groups of the fiber sequence

$$L_{bir}\tau^\sigma_{>i}\mathcal{X} \to L_{bir}\mathcal{X} \to L_{bir}\tau^\sigma_{\leq i}\mathcal{X},$$

this yields $L_{bir}\tau^\sigma_{\leq i}\mathcal{X} \simeq \tau^\sigma_{\leq i}L_{bir}\mathcal{X}$.

On the other hand, we compare the fiber sequence in the first line with the fiber sequence

$$\tau^\sigma_{>i}L_{bir}\mathcal{X} \to L_{bir}\mathcal{X} \to \tau^\sigma_{\leq i}L_{bir}\mathcal{X}$$

given by the n-th slice of the Postnikov tower of $L_{bir}\mathcal{X}$.

$$\begin{array}{ccccc} L_{bir}\tau^\sigma_{>i}\mathcal{X} & \longrightarrow & L_{bir}\mathcal{X} & \longrightarrow & L_{bir}\tau^\sigma_{\leq i}\mathcal{X} \\ \downarrow & & \downarrow & & \downarrow \\ \tau^\sigma_{>i}L_{bir}\mathcal{X} & \longrightarrow & L_{bir}\mathcal{X} & \longrightarrow & \tau^\sigma_{\leq i}L_{bir}\mathcal{X} \end{array}$$

Since both the middle and right vertical morphisms are equivalences, we deduce that the left vertical map is also an equivalence. □

Corollary 3.6.5. *Let $\mathcal{X} \in \mathcal{P}_\sigma(S)_{\geq 1}$ be pointed, with a σ-localy simply connected cover $\widetilde{\mathcal{X}}$. If $\pi^\sigma_1\mathcal{X}$ is birational, then $L_{bir}\widetilde{\mathcal{X}} \to L_{bir}\mathcal{X}$ is a birationally simply connected covering. It follows that there is an isomorphism*

$$\pi_1\mathcal{X} \simeq \pi^b_1\mathcal{X} \simeq Aut_{L_{bir}\mathcal{X}}(L_{bir}\widetilde{\mathcal{X}}).$$

Proof. Applying Proposition 3.6.4 with $n = 1$ it follows that

$$L_{bir}\tau^{nis}_{>1}\mathcal{X} \to \tau^{nis}_{>1}L_{bir}\mathcal{X}$$

is an equivalence. The claim follows from the fact that for a pointed σ-locally connected space $\mathcal{Y}$, $\tau^\sigma_{>1}\mathcal{Y}$ is its unique pointed simply connected cover. □

Remark 3.6.6. In principle, similar to Morel's result in his book [Mor12, Theorem 7.8], one would like a statement about the existence of birational covering spaces over birationally connected spaces, and that too without the birationality assumption of π_1. Unfortunately, because L_{bir} is not locally cartesian, we are unable to establish this in such generality. However, Corollary 3.2.15 suggests that this is possible over $\mathbb{A}^1$-connected objects. This will require categorical discussions of covering spaces, which we will develop in an upcoming work [Mai26c, §2.4]. As a consequence, we will be able to construct the desired birational result over $\mathbb{A}^1$-connected spaces [Mai26c, Theorem 4.4.14]. In the same paper, we will also reprove some versions of Morel's results on $\mathbb{A}^1$ covering spaces, adapted to the ∞-categorical language (see §3.4 in *loc.cit.*).

Theorem 3.6.7. *Let $\mathcal{X}$ be a pointed σ-locally 0-connected S-space such that, for some $n \geq 0$, the space $\tau^{\sigma}_{\leq n}\mathcal{X}$ is birational (which occurs precisely when $\pi^{\sigma}_i\mathcal{X}$ are birational local for all $i \leq n$; see Corollary 2.1.9). Then the morphism $\pi^{\sigma}_{n+1}\mathcal{X} \to \pi^{b}_{n+1}\mathcal{E}$ is universal among morphisms to a birational sheaf of groups when $n = 0$ and to a birational sheaf of abelian groups when $n \geq 1$.*

Proof. Let B be a birational sheaf of (abelian if $n \geq 1$) groups. Since $\tau^{\sigma}_{>n}\mathcal{X}$ is σ-locally n-connected by definition, it admits an $(n+1)$-fold delooping. In other words, we have an equivalence $\tau^{\sigma}_{>n}\mathcal{X} \simeq \mathrm{B}^{n+1}_{\sigma}\Omega^{n+1}\tau^{\sigma}_{>n}\mathcal{X}$. But then

$$\begin{aligned}
\mathrm{Map}_{\mathcal{P}_\sigma(S)}(\tau^{\sigma}_{>n}\mathcal{X}, \mathbf{B}^{n+1}_{\sigma}B) &\simeq \mathcal{E}_{n+1}\text{-}\mathrm{Mon}_{\mathcal{P}_\sigma(Sm_S)}(\Omega^{n+1}\tau^{\sigma}_{>n}\mathcal{X}, B) && (6)\\
&\cong (n+1)\text{-}Grp^{\sigma}_{S}(\pi_0\Omega^{n+1}\tau^{\sigma}_{>n}\mathcal{X}, B) \text{ [since } B \text{ is discrete]} && (7)\\
&\cong (n+1)\text{-}Grp^{\sigma}_{S}(\pi^{\sigma}_{n+1}\mathcal{X}, B) && (8)
\end{aligned}$$

On the other hand, since the space $\mathrm{B}^{n+1}_{\sigma}B = \mathrm{K}(B, n+1)$ is birational local (see Corollary 3.2.6), the starting infinity groupoid is identical to

$$\mathrm{Map}_{\mathcal{H}^b(S)}(L_{bir}\tau^{\sigma}_{>n}\mathcal{X}, \mathbf{B}^{n+1}_{\sigma}B).$$

By Proposition 3.6.4, this mapping space can be further identified with

$$\mathrm{Map}_{\mathcal{P}_\sigma(S)}(\tau^{\sigma}_{>n}L_{bir}\mathcal{X}, \mathrm{B}^{n+1}_{\sigma}B).$$

By the same line of identifications as in the above computation (6-8), this last term is equivalent to

$$(n+1)\text{-}Grp^{\sigma}_{S}(\pi^{\sigma}_{n+1}L_{bir}\mathcal{X}, B).$$

Again, because $L_{bir}\mathcal{X}$ is birational $\pi_{n+1}L_{bir}\mathcal{X} \simeq \pi^{\sigma}_{n+1}L_{bir}\mathcal{X}$. Thus,

$$(n+1)\text{-}Grp^{\sigma}(\pi^{\sigma}_{n+1}L_{bir}\mathcal{X}, B) \cong (n+1)\text{-}Grp^{b}(\pi^{b}_{n+1}\mathcal{X}, B).$$

One may now use the Yoneda lemma to conclude. (here, $(n+1)$-Grp^{σ}_{S} is the category of σ-sheaves of $(n+1)$-group objects; when $n = 0$, this is simply Grp^{σ}_{S}, whereas for $n \geq 1$, an Eckmann-Hilton argument shows that these are all equivalent to Ab^{σ}_{S}.) □

The π_0 case does not require any condition:

Proposition 3.6.8. *For any $\mathcal{X} \in \mathcal{P}(S)$, the natural map $\pi^{\sigma}_0\mathcal{X} \to \pi^{b}_0\mathcal{X}$ is the universal morphism to a birational (pre)sheaf of sets. In particular, replacing $\mathcal{X}$ by $L_{mot}\mathcal{X}$ yields that $\pi^{\mathbb{A}^1}_0\mathcal{X} \to \pi^{b}_0\mathcal{X}$ is a birationalization.*

Proof. Since a birational sheaf of sets B is discrete, we compute

$$\mathrm{Map}_{\mathcal{P}(Sm_S)}(\mathcal{X}, B) \simeq PSh_S(\pi_0\mathcal{X}, B).$$

Again, since B is birational as a space, the left-hand side is

$$\mathrm{Map}_{\mathcal{H}^b(S)}(L_{bir}\mathcal{X}, B) \simeq \mathrm{Map}_{\mathcal{P}(S)}(L_{bir}\mathcal{X}, B) \simeq PSh_S(\pi_0L_{bir}\mathcal{X}, B) = Shv^{b}_{S}(\pi^{b}_0\mathcal{X}, B).$$

These equivalences induce a canonical identification

$$PSh_S(\pi_0\mathcal{X}, B) \cong Shv_S^b(\pi_0^b\mathcal{X}, B).$$

Formally, this implies that $\pi_0^b\mathcal{X}$ is the birationalization of $\pi_0\mathcal{X}$. Since birational presheaves are σ-sheaves, this immediately guarantees (by the universal property of sheafification) that $\pi_0^b\mathcal{X}$ is the birationalization of $\pi_0^\sigma\mathcal{X}$ as well. For the last statement, use $\sigma = nis$. □

Remark 3.6.9. For a discrete presheaf of sets, this implies that $F \to \pi_0^b F$ provides a left adjoint to the inclusion $Shv_S^b \hookrightarrow Shv_S^\sigma$ (which has been studied over fields earlier in [Shi23], where it is denoted π_0^{br}). So the observations of the above proposition can be restated as

$$\pi_0^b\mathcal{X} \simeq \pi_0^b\pi_0\mathcal{X} \simeq \pi_0^b\pi_0^{nis}\mathcal{X} \simeq \pi_0^b\pi_0^{\mathbb{A}^1}\mathcal{X}.$$

Moreover, since both π_0 and L_{bir} preserve finite products [Theorem 3.2.2], we know that $\pi_0^b = \pi_0 L_{bir}$ also preserves them. So in particular, π_0^b preserves group structures. Therefore, the conclusion of Theorem 3.6.7 can be restated as

$$\pi_{n+1}^b\mathcal{X} \cong \pi_0^b\pi_{n+1}^\sigma\mathcal{X}.$$

Corollary 3.6.10. *Let $\mathcal{M}$ be a commutative H-space. Then $\pi_0\mathcal{M} \to \pi_0^b\mathcal{M}$ is a surjection of (pre)sheaves.*

Proof. From Proposition 3.6.8 we know that $\pi_0\mathcal{M} \to \pi_0^b\mathcal{M}$ is an epimorphism in $CMon^b$. But $CMon_S^b \subset CMon_S$ is exact (the same proof as in Corollary 3.2.10 applies; the existence of an inverse in the cited result is irrelevent). So $\pi_0\mathcal{M} \to \pi_0^b\mathcal{M}$ is an epimorphism in PSh_S. □

Remark 3.6.11. This is a version of the $\mathbb{A}^1$ motivic result [MV99, Corollary 2.1.3] which works for all S-spaces. Birational locally this is certainly not true for all spaces as Remark 3.5.6 shows.

Corollary 3.6.12. *Suppose $\mathcal{X}$ is a (motivically) pointed $\mathbb{A}^1$-n-connected S-space. Then the canonical map $\pi_{n+1}^{\mathbb{A}^1}\mathcal{X} \to \pi_{n+1}^b\mathcal{X}$ is universal among maps to a birational sheaf of sets when $n = -1$; to a birational sheaf of groups when $n = 0$; and to a birational sheaf of abelian groups when $n \geq 1$; i.e.,*

$$\pi_{n+1}^b\mathcal{X} \simeq \pi_0^b(\pi_{n+1}^{\mathbb{A}^1}\mathcal{X}).$$

Proof. When $n = -1$, this is Proposition 3.6.8. For $n \geq 0$, apply Theorem 3.6.7 to the space $\mathcal{Y} := L_{mot}\mathcal{X}$ with the Nisnevich topology. By the given condition, $\tau_{\leq n}^{nis}\mathcal{Y} \simeq pt$ is trivially birational. It follows that $\pi_{n+1}^{nis}\mathcal{Y} \to \pi_{n+1}^b\mathcal{Y}$ is the universal birational sheaf. But this map is identical to the map in question, since $L_{bir}L_{mot} \simeq L_{bir}$. The equivalence $\pi_{n+1}^b\mathcal{X} \simeq \pi_0^b(\pi_{n+1}^{\mathbb{A}^1}\mathcal{X})$ (as groups) follows from Remark 3.6.9. □

Remark 3.6.13. When the base is the spectrum of a perfect field k and $n \geq 0$, the above condition can be restated, using Lemma 3.3.9, as saying that the map $\pi_{n+1}^{\mathbb{A}^1}\mathcal{X} \to \pi_{n+1}^b\mathcal{X}$ is universal among maps to a Nisnevich sheaf of (abelian when $n \geq 1$) groups that is (strictly when $n \geq 1$) strongly-$\mathbb{A}^1$-invariant and has trivial $\mathbb{G}_m$-contraction.

Computing birational path component π_0^b:

Therefore, the transition from motivic homotopy (groups) to birational motivic homotopy (groups) is, to some extent, controlled by π_0^b of discrete S-spaces. We want to observe that in sufficiently nice scenarios, this behaves quite well. In fact, it turns out that computing the

birational path component of any presheaf of spaces is easier than computing its $\mathbb{A}^1$ counterpart. To see this, recall that by Lemma 3.1.6, for $\mathcal{X} \in \mathcal{P}_\Sigma(S)$ we may write

$$L_{bir}\mathcal{X} \simeq \operatorname*{colim}_{\substack{X \to \mathcal{X} \\ X \in Sm_S}} L_{bir}X$$

where the colimit is computed in $\mathcal{P}(S)$ and is taken over the diagram of sections of $\mathcal{X}$ over $X/S \in Sm_S$ having property P. Since $\pi_0 : \mathcal{P}(S) \to PSh_S$ preserves colimits, it follows that

$$\pi_0^b\mathcal{X} = \pi_0 L_{bir}\mathcal{X} \simeq \operatorname*{colim}_{\substack{X \to \mathcal{X} \\ X/S \in Sm_S}} \pi_0^b X \tag{9}$$

where the colimit is computed in the category of presheaves of sets on Sm_S. Thus, to compute $\pi_0^b\mathcal{X}$ for any $\mathcal{X}$, it suffices to know $\pi_0^b X$ for $X \in Sm_S$ having property P.

For example, the referred theorem shows that, when the base is a field k of characteristic zero, P can always be taken to be the property that X/k is smooth and proper. Therefore, in characteristic 0, to compute π_0^b of any space, it suffices to know $\pi_0^b X$ for every smooth proper scheme X/k.

The computation of these sheaves, however, can be performed over any field and for all proper schemes. So, let us fix an arbitrary base field k. Given a proper scheme X/k, there is a birational sheaf, denoted $\pi_0^{b\mathbb{A}^1}X$, whose sections over a finitely generated separable field extension L/k is given by the set of (Manin's) R-equivalence classes of L-points of X [AM11, §6]. The method used to prove its existence is as follows. The authors first introduce the category $Shv_k^{b\mathbb{A}^1}$, whose objects are birational sheaves that are $\mathbb{A}^1$-invariant. Then they prove that the restriction functor $Shv_k^{b\mathbb{A}^1} \to \mathcal{F}_k^r\text{-}Sets$, given by evaluation at the generic point, is fully faithful, with essential image consisting of sheaflike transcendental invariants (which they call $\mathbb{A}^1$-invariant) data [AM11, Theorem 6.1.7]. Although it is not difficult do see that,

Lemma 3.6.14. *For a field k, the inclusion $Shv_k^{b\mathbb{A}^1} \subset Shv_k^b$ is an equality, and the functor $Snv_k^b \to \mathcal{F}_k^r$-Sets given by evaluation at the generic point is fully faithful, with essential image consisting of sheaf-like transcendental invariant data (see [AM11, §6] for definition).*

Proof. This is because birational presheaves of sets are automatically $\mathbb{A}^1$-invariant (Corollary 2.2.7). □

Finally, the authors of *loc.cit.* prove that, for a proper variety X/k, the assignment $L \mapsto X(L)/\sim$ of naive $\mathbb{A}^1$-homotopy classes of L-points of X is a sheaflike transcendental invariant $\mathcal{F}_k^r$-data (see Theorem 6.2.1 of *loc.cit.*). We claim that this is exactly the candidate $\pi_0^b X$ we were looking for. This result was obtained for **smooth proper varieties** in [CR24, Theorem 3.4] using Kahn-Sujhata's localized category $S_b^{-1}Sm_k$. The proof below is new. It is formal and does not require $S_b^{-1}Sm_k$, and thus works for all proper schemes:

Theorem 3.6.15 (this is a generalization of [CR24, Theorem 3.4] and [CK23, Proposition 1.9.] to all proper schemes)**.** *Let k be a field and X a proper scheme over k. Then there is a canonical isomorphism $\pi_0^b X \simeq \pi_0^{b\mathbb{A}^1}X$ of presheaves.*

Proof. By Proposition 3.6.8, we know that $X \cong \pi_0 X \to \pi_0^b X$ provides the birationalization of X. But by [AM11, Remark 6.2.2], for any proper scheme X, the canonical morphism $X \to \pi_0^{b\mathbb{A}^1}X$ is the universal object of $Shv_k^{b\mathbb{A}^1}$ associated to X. Because, $Shv_k^{b\mathbb{A}^1} = Shv_k^b$, $X \to \pi_0^{b\mathbb{A}^1}X$ is in fact, the universal birationalization. By comparing the universal properties, one immediately obtains the canonical isomorphism $\pi_0^b X \xrightarrow[\sim]{} \pi_0^{b\mathbb{A}^1}X$. □

Corollary 3.6.16. *For $X \in \mathrm{Prop}_k$, the canonical maps*

$$X \to \pi_0^{b\mathbb{A}^1} X \text{ and } \pi_0^{\mathbb{A}^1} X \to \pi_0^{b\mathbb{A}^1} X$$

[AM11, Proposition 6.2.6] *are initial among maps to a birational sheaf of sets.*

Proof. Combine Proposition 3.6.8 and Theorem 3.6.15. □

Corollary 3.6.17 (this is a generalization of [Koi22, Theorem 2.4])**.** *The assignment*

$$\pi_0^{b\mathbb{A}^1}(-) : \mathrm{SmProp}_k \to Sh_k$$

is a stable birational invariant, i.e., if X and Y are stably birational smooth proper schemes, then there is an isomorphism of sheaves $\pi_0^{b\mathbb{A}^1} X \simeq \pi_0^{b\mathbb{A}^1} Y$.

Proof. Note that the assignment

$$\pi_0^b(-)_{|\mathrm{Sm}_k} : \mathrm{Sm}_k \to Sh_k$$
$$X \mapsto \pi_0^b(X) = \pi_0 L_{bir} X$$

by construction, is a stable birational invariant. The claim then follows from the theorem above, Theorem 3.6.15. □

Here is a purely motivic consequence ($\mathbb{H}_0^{\mathbb{A}^1}$ stands for 0-th $\mathbb{A}^1$-homology sheaf, as introduced by Morel [Mor12, Definition 6.29]):

Corollary 3.6.18. *The assignment*

$$\mathbb{H}_0^{\mathbb{A}^1}(-) : \mathrm{SmProp}_k \to Ab_k^{\mathbb{A}^1}$$

factors through Ab_k^b and is itself a stable birational invariant.

Proof. By [Koi22, Theorem 4.5], we know that if X is smooth and proper, then

$$\mathbb{H}_0^{\mathbb{A}^1}(X) \simeq \mathbb{Z}(\pi_0^{b\mathbb{A}^1} X).$$

Thus, the first claim follows from the fact that $\mathbb{Z}(-)$ preserves birational locality, while the second follows from the corollary just above. □

Corollary 3.6.19. *For a smooth proper scheme X/k, the canonical maps $X \to \mathbb{H}_0^{\mathbb{A}^1} X$ and $\pi_0^{\mathbb{A}^1} X \to \mathbb{H}_0^{\mathbb{A}^1} X$ are initial among maps to a birational sheaf of abelian groups.*

Proof. This is immediate from the identity $\mathbb{H}_0^{\mathbb{A}^1}(X) \simeq \mathbb{Z}(\pi_0^b X)$, as noted above, and from Corollary 3.6.16. □

Let us now go back to what we started with in this 3.6. There we saw that when k is a field of characteristic 0, the birational path component of any $\mathcal{X} \in \mathcal{P}_\Sigma(k)$ can be computed as in Equation (9). The main point of this small subsection is that using Theorem 3.6.15, we can now rewrite the above-mentioned formula by replacing $\pi_0^b X$ with $\pi_0^{b\mathbb{A}^1} X$:

$$\pi_0^b \mathcal{X} = \pi_0 L_{bir} \mathcal{X} \simeq \operatorname*{colim}_{\substack{\bar{X} \hookleftarrow X \to \mathcal{X} \\ X \in Sm_k}} \pi_0^{b\mathbb{A}^1} X' \tag{10}$$

where the colimit is taken over spans with $X \hookrightarrow \bar{X}$ a dense open immersion into smooth proper $\bar{X}$. The benefit of this presentation is that, as explained earlier, the candidate $\pi_0^{b\mathbb{A}^1} X$ has a purely geometric description: generically, this is the sheaf of (Manin's) R-equivalent points of X.

Remark 3.6.20. Recall that the motivic homotopy category is also generated by sifted colimits of motivic models of smooth (affine) schemes. However, the tricks used in our birational case do not quite apply to the ordinary motivic case. The problem is multifold:

(1) First, since the motivic homotopy category $\mathcal{H}^{\mathbb{A}^1}(S)$ is not closed under sifted colimits in $\mathcal{P}_{nis}(S)$ [Remark 3.1.2], the above-mentioned colimits are computed only in $\mathcal{H}^{\mathbb{A}^1}(S)$, not in $\mathcal{P}_{nis}(S)$. Therefore, π_0^{nis} need not preserve these colimits.
(2) On the other hand, since $\pi_0^{\mathbb{A}^1}$ is not the internal 0-truncation of $\mathcal{H}^{\mathbb{A}^1}(S)$ [Ayo23], applying even $\pi_0^{\mathbb{A}^1}$ might not help either.
(3) Finally, even if one could manage the above problems, in a sufficiently nice setup, computing $\pi_0^{\mathbb{A}^1}$ is known to be extremely difficult, even for schemes. In fact, even over fields, there is no class of schemes known to date that generates the motivic homotopy category and for which $\pi_0^{\mathbb{A}^1}$ is completely known.

The observation of Theorem 3.6.15 makes π_0^b into a powerful device. So much so that it can detect connectivity of 'proper' motivic spaces (the last term means an object of the subcategory of $\mathcal{H}^{\mathbb{A}^1}(k)$ generated under filtered colimits by proper schemes).

Proposition 3.6.21. *Let $\mathcal{H}^{\mathbb{A}^1}_{prop}(k)$ be the full subcategory of $\mathcal{H}^{\mathbb{A}^1}(k)$ generated under filtered colimits by proper schemes. Then the restriction of the functor $L_{bir} : \mathcal{H}^{\mathbb{A}^1}(k) \to \mathcal{H}^b(k)$ to $\mathcal{H}^{\mathbb{A}^1}_{prop}(k)$ preserves and detects connectivity.*

Proof. Preservation of connectivity follows from Corollary 3.5.9(2).

To address detection, let $\mathcal{X}$ be a motivic local space, given by the filtered colimit

$$\operatorname*{colim}_{X \in \mathrm{Prop}/\mathcal{X}} L_{mot}(X)$$

in $\mathcal{H}^{\mathbb{A}^1}(k)$. Since $\mathcal{H}^{\mathbb{A}^1}(k) \subset \mathcal{P}_{nis}(k)$ is closed under filtered colimits, we see that this colimit can be computed in $\mathcal{P}_{nis}(k)$. Since $L_{bir} : \mathcal{H}^{\mathbb{A}^1}(k) \to \mathcal{P}_{nis}(k)$ preserves filtered colimits (indeed, $L_{bir} : \mathcal{H}^{\mathbb{A}^1}(k) \to \mathcal{H}^b(k)$ is a left adjoint and $\mathcal{H}^b(k) \subset \mathcal{P}_{nis}(k)$ is closed under filtered colimits (see Theorem 3.1.1), we deduce that the morphism $\mathcal{X} \to L_{bir}\mathcal{X}$ can be identified with the colimit

$$\operatorname*{colim}_{X \in \mathrm{Prop}/\mathcal{X}} (L_{mot}X \to L_{bir}X)$$

in $\mathcal{P}_{nis}(k)$. Applying π_0^{nis}, we find that $\pi_0^{\mathbb{A}^1}\mathcal{X} \to \pi_0^b\mathcal{X}$ is the colimit

$$\operatorname*{colim}_{X \in \mathrm{Prop}/\mathcal{X}} (\pi_0^{\mathbb{A}^1}X \to \pi_0^bX).$$

Since every X in the colimit above is proper, Theorem 3.6.15 implies that each term in the colimit is an isomorphism over $\mathcal{F}_k$ [Aso13, Proposition 6.2.6], and hence so is $\pi_0^{\mathbb{A}^1}\mathcal{X} \to \pi_0^b\mathcal{X}$. Thus, when $\pi_0^b\mathcal{X}$ is trivial, we have that $\pi_0^{\mathbb{A}^1}\mathcal{X}$ is trivial over $\mathcal{F}_k$. Since triviality of $\pi_0^{\mathbb{A}^1}\mathcal{X}$ can be detected over $\mathcal{F}_k$ [Bac24, Corollary 2.10.], we are done. □

Corollary 3.6.22. *An (ind-)proper scheme over a field k is $\mathbb{A}^1$-connected if and only if it is birationally connected.*

Remark 3.6.23. The counterexample in Remark 2.2.10 is a threefold. For (proper) schemes of dimension ≤ 2, however, this is not possible. That is, birationally contractible proper surfaces are necessarily rational, at least when k is algebraically closed of characteristic zero. In fact, any birationally connected surface is rational. Indeed, by the above corollary, such varieties are $\mathbb{A}^1$-connected. But in characteristic zero, $\mathbb{A}^1$-connected schemes of dimension ≤ 2 are necessarily rational [AM11, Corollary 2.4.7].

Remark 3.6.24. The above proposition and its corollary are clearly false on non-proper spaces. Take, for example, $\mathbb{G}_m$. This space is $\mathbb{A}^1$-disconnected despite being birationally connected (in fact, birationally contractible).

Remark 3.6.25. These might also fail for higher connectivity. For example, $\mathbb{P}^2_k$ is birationally contractible over k, but has non trivial $\mathbb{A}^1$-fundamental group: $\pi_1^{\mathbb{A}^1}\mathbb{P}^2 \cong \mathbb{G}_m$ [Mor12, Theorem 7.13].

We end this section with a birational analog of the Freudenthal Suspension theorem, which follows essentially from Theorem 3.6.7 and the methods of [Mor12, Theorem 6.61].

Theorem 3.6.26 (Birational Freudenthal suspension theorem)**.** *Assume that S is Qcqs and that $\mathcal{E}$ is a pointed S-space that is birationally n-connected for some $n \geq 1$. Then:*

(1) The canonical adjoint unit map $\eta^b_{\mathcal{E}} : L_{bir}\mathcal{E} \to \Omega L_{bir}\Sigma\mathcal{E}$ has $2n$-connected fiber.
(2) The adjoint unit map $\eta_{\mathcal{E}} : \mathcal{E} \to \Omega\Sigma\mathcal{E}$ has a $2n$-connected birational fiber. If $\mathcal{E}$ is connected, then, in fact, the ordinary fiber of $\eta_{\mathcal{E}}$ is birationally $2n$-connected.
(3) The canonical map $L_{bir}\mathcal{E} \to \Omega^\infty\Sigma^\infty_{bir}\mathcal{E}$ has $2n$-connected fiber. Here $\Sigma_{bir} := L_{bir}\Sigma$ is the internal suspension of $\mathcal{H}^b(S)$; equivalently,

$$\Sigma_{bir}\mathcal{X} = \operatorname{colim}^{\mathcal{H}^b(S)}(pt \leftarrow L_{bir}\mathcal{X} \to pt).$$

Proof. 1. We are given that $L_{bir}\mathcal{E}$ is n-connected. The classical Freudenthal suspension theorem, applied to this $2n$-connected space $L_{bir}\mathcal{E}$, yields that the fiber of the unit $\eta_{L_{bir}\mathcal{E}} : L_{bir}\mathcal{E} \to \Omega\Sigma L_{bir}\mathcal{E}$ is $2n$-connected. That is, the morphism

$$f : \pi^b_{2n+1}\mathcal{E} = \pi_{2n+1}L_{bir}\mathcal{E} \to \pi_{2n+2}\Sigma L_{bir}\mathcal{E}$$

is an epimorphism of presheaves, and

$$\tau_{\leq 2n}L_{bir}\mathcal{E} \xrightarrow{\tau_{\leq 2n}\eta_{L_{bir}\mathcal{E}}} \tau_{\leq 2n}\Omega\Sigma L_{bir}\mathcal{E} \simeq \Omega\tau_{\leq 2n+1}\Sigma L_{bir}\mathcal{E}$$

is an equivalence. Since the space $\tau_{\leq 2n+1}\Sigma L_{bir}\mathcal{E}$ is pointed and connected, this equivalence implies

$$\tau_{\leq 2n+1}\Sigma L_{bir}\mathcal{E} \simeq \mathrm{B}\Omega\tau_{\leq 2n+1}\Sigma L_{bir}\mathcal{E} \simeq \mathrm{B}\tau_{\leq 2n}L_{bir}\mathcal{E}.$$

Since $\mathrm{B}\tau_{\leq 2n}L_{bir}\mathcal{E}$ is birational local by Lemma 3.3.4 and Lemma 3.2.3(2), it follows that so is $\tau_{\leq 2n+1}\Sigma L_{bir}\mathcal{E}$. But then

$$\tau_{\leq 2n+1}\Sigma L_{bir}\mathcal{E} \simeq L_{bir}\tau_{\leq 2n+1}\Sigma L_{bir}\mathcal{E} \simeq \tau_{\leq 2n+1}L_{bir}\Sigma L_{bir}\mathcal{E} \simeq \tau_{\leq 2n+1}L_{bir}\Sigma\mathcal{E}$$

(for the second equivalence use Proposition 3.6.4). Now, applying Ω to the above equivalence, we deduce that

$$\Omega\tau_{\leq 2n+1}\Sigma L_{bir}\mathcal{E} \simeq \Omega\tau_{\leq 2n+1}L_{bir}\Sigma\mathcal{E} \simeq \tau_{\leq 2n}\Omega L_{bir}\Sigma\mathcal{E}.$$

Composing with the starting (given) equivalence $\tau_{\leq 2n}L_{bir}\mathcal{E} \simeq \Omega\tau_{\leq 2n+1}\Sigma L_{bir}\mathcal{E}$ this gives us an equivalence:

$$\tau_{\leq 2n}L_{bir}\mathcal{E} \simeq \tau_{\leq 2n}\Omega L_{bir}\Sigma\mathcal{E}.$$

On the other hand, birational locality of $\tau_{\leq 2n+1}\Sigma L_{bir}\mathcal{E}$ yields, by Theorem 3.6.7, that $\pi_{2n+2}\Sigma L_{bir}\mathcal{E} \to \pi^b_{2n+2}\Sigma L_{bir}\mathcal{E}$ is the universal one to a birational abelian group sheaf. But

$$\pi^b_i\Sigma L_{bir}\mathcal{E} = \pi_i L_{bir}\Sigma L_{bir}\mathcal{E} \simeq \pi_i L_{bir}\Sigma\mathcal{E} \simeq \pi^b_i\Sigma\mathcal{E}.$$

Therefore the morphism $\pi_{2n+2}\Sigma L_{bir}\mathcal{E} \to \pi^b_{2n+2}\Sigma\mathcal{E}$ is the universal morphism to a birational abelian group sheaf. Hence, the composition

$$\pi^b_{2n+1}\mathcal{E} \xrightarrow{f} \pi_{2n+2}\Sigma L_{bir}\mathcal{E} \to \pi^b_{2n+2}\Sigma\mathcal{E}$$

is an epimorphism in Ab^b_S (indeed, f is given to be an epimorphism of presheaves and thus an epimorphism on birationalization, while the second one is an isomorphism on birationalization). Since $Ab^b_S \subset PAb_S$ is exact (use Corollary 3.2.10 for the trivial topology), we deduce that

$$\pi_{2n+1} L_{bir}\mathcal{E} \to \pi_{2n+1}\Omega L_{bir}\Sigma\mathcal{E}$$

is an epimorphism in PAb_S.

We have thus shown that the morphism $\eta^b_{\mathcal{E}} : L_{bir}\mathcal{E} \to \Omega L_{bir}\Sigma\mathcal{E}$ is an equivalence under $\tau_{\leq 2n}$ and induces a surjection on π_{2n+1}, which is equivalent to saying that it has $2n$-connected fiber.

2. Since $\Sigma\mathcal{E}$ is connected, Theorem 3.2.12 implies that the codomain of the morphism in (1) is equivalent to $L_{bir}\Omega\Sigma\mathcal{E}$. Hence, the first statement of (2) is a restatement of (1). When $\mathcal{E}$ is furthermore connected, so that $\Omega\Sigma\mathcal{E}$ is connected, Lemma 3.2.11 implies that $L_{bir} fib(\eta_{\mathcal{E}}) \simeq fib(L_{bir}\eta_{\mathcal{E}})$, and the second line of (2) follows from the first.

3. The morphism in the third statement is the colimit of the following diagram:

$$L_{bir}\mathcal{E} \xrightarrow{\eta_{\mathcal{E}}} \Omega\Sigma_{bir}\mathcal{E} \xrightarrow{\Omega\eta_{\Sigma_{bir}\mathcal{E}}} \Omega^2\Sigma^2_{bir}\mathcal{E} \longrightarrow \cdots$$

Inductively, it follows from statement (1) that each morphism in the above diagram is $2n$ conneceted. Hence, so is the colimit (use [Lur09, Proposition 6.5.1.12]). □

4. Examples

4.1. **Examples of Birational Motivic spaces.** In this section, let k be a field.

Birationally rigid schemes[1]

Definition 4.1.1. *We say that a scheme T/k is birationally rigid if, for every dense open immersion $U \subset X$ of smooth schemes, the restriction maps $\mathrm{Sch}_k(X,T) \to \mathrm{Sch}_k(U,T)$ are isomorphisms, i.e., if it admits no nontrivial rational maps. Since schemes are Nisnevich sheaves, they are precisely the schemes that are birational local (see Definition 2.0.2).*

Example 4.1.2. A zero-dimensional (not necessarily smooth) locally Noetherian scheme over k is birationally rigid, since its underlying topological space is discrete [Sta26, Lemma 0AAX].

Example 4.1.3 (Abelian varieties)**.** The standard example of a Birationally rigid variety is an abelian variety [Mil86, Theorem 3.2]. In particular, elliptic curves over k are birationally rigid.

Since birational local spaces are already $\mathbb{P}^1_k$-local (Corollary 2.2.9(1)), the following is immediate:

Proposition 4.1.4. *A birationally rigid scheme contains no rational curve.*

In particular, a nontrivial rationally connected variety is never birationally rigid. The converse holds for proper schemes:

Proposition 4.1.5. *A proper variety with no rational curves is birationally rigid.*

Proof. Indeed, suppose T is a proper variety over k with no rational curve, and we are given a morphism $f : U \to T$ with $U \in Sm_k$. Then by [Deb01, Corollary 1.44], f extends along the dense open immersion $U \hookrightarrow X \in Sm_k$.

To elaborate, consider the graph of f, given by $\Gamma_f \subset U \times T \subset X \times T$. Let $\bar{\Gamma}_f \subset X \times T$ be the closure of Γ_f in $X \times T$, and let $\phi : \bar{\Gamma}_f \to X$ be the corresponding projection map, which is

[1]This should not be confused with the notion of birational rigidity used in classical birational geometry [Puk13]. In fact, the geometric concept is quite the opposite. For instance, being rationally connected, Fano varieties are never rigid in our sense, yet many Fano varieties are rigid in the other sense (see [Che05]).

proper since T is so. Since $U \hookrightarrow X$ is dense and $U \cong \Gamma_f$, we get that ϕ is birational. Since X is smooth, the irreducible components of the exceptional fiber are ruled (by Proposition 1.43 in *loc.cit.*). Since T contains no rational curve, each fiber of ϕ must map to a single point in T. Because ϕ is surjective by construction, we get an extension to all of X. □

Corollary 4.1.6. *A proper scheme over k is birationally rigid if and only if it is rationally rigid (i.e., L_{rat} local), if and only if it is $\mathbb{P}^1$ rigid, if and only if it is $\mathbb{A}^1$ rigid.*

Proof. For a proper scheme, the equivalence of $\mathbb{P}^1$-rigidity and $\mathbb{A}^1$-rigidity follows from the valuative criterion of properness. By Corollary 2.2.9, birational local schemes are rational local. Since $\mathbb{A}^1$ is a subscheme of $\mathbb{P}^1$, in general, $\mathbb{A}^1$-rigid schemes are $\mathbb{P}^1$-rigid (in fact, this holds for all nisnevich sheaves of sets, see [Mai26b, Example 2.2.6]). The remaining implication follows from the result above (note that the absence of rational curves is equivalent to $\mathbb{P}^1$-rigidity). □

Example 4.1.7. For example, smooth proper curves of genus ≥ 1 are birationally rigid (this appears in [Lev10, page 399]).

Example 4.1.8. The limit of a (finite) diagram consisting of birationally rigid schemes is birationally rigid. For example, a product of smooth proper curves of positive genus is birationally rigid.

Path components of standard correspondences

Let Y be a smooth proper variety over a field k.

Example 4.1.9 (Chow groups of zero cycles of smooth proper varieties)**.** The presheaf with transfers associated to the naive $\mathbb{A}^1$-homotopy quotient of a proper variety Y, i.e.,

$$\mathrm{H}_0^{\mathbb{Z}}(Y) := \pi_0^{nis} L_{\mathbb{A}^1} \mathbb{Z}^{tr}(Y),$$

is a birational sheaf. Indeed, by [KS17, Theorem 3.1.2]

$$\pi_0^{nis} L_{\mathbb{A}^1} \mathbb{Z}^{tr}(Y) \cong \mathrm{CH}_0(Y_{K-}),$$

where the right hand side is clearly birational (note that the only thing that requires properness of Y is in prividing the right hand side with a presheaf structure, otherwise $X \mapsto \mathrm{CH}_0(Y_{KX})$ is not functorial on all morphisms of Sm_k).

Remark 4.1.10. A similar claim (and consequently an appropriate isomorphism as mentioned earlier) does not apply to Milnor-Witt or framed correspondences. Specifically, when $Y = Speck$, we observe

$$\pi_0^{nis} L_{\mathbb{A}^1} \mathbb{Z}^{fr}(k) \simeq \pi_0^{nis} L_{\mathbb{A}^1} \tilde{\mathbb{Z}}(k) \simeq \underline{\mathrm{GW}}.$$

(The first equivalence follows from [AN19, Theorem 8.8].) However, it is well known that the unramified sheaf $\underline{\mathrm{GW}}$ is not birational local (for example, using the Rost-Schmid complex, and $\mathbb{A}^1$-invariance, it is clear that

$$\underline{\mathrm{GW}}(\mathbb{G}_{m,k}) \simeq GW(k) \oplus W(k)$$

and $W(k)$ is often nontrivial).

More discrete birational sheaves

Example 4.1.11. Over geometrically unibranch schemes, every constant sheaf of sets is birationally local. Indeed, in this case, the constant sheaf associated to a set F is given by $U \mapsto \sqcup_{U^{(0)}} F$, since the connected components of unibranch schemes are irreducible.

Example 4.1.12. As noted earlier, for a field k, [AM11, Theorem 6.1.7] states that any $\mathcal{F}_k^r - set$ that is sheaflike and transcendental invariant [Definition 6.1.6 of *loc.cit.*] can be extended to a birational sheaf on Sm_k.

Non-discrete birational local spaces

Example 4.1.13. It follows from Example 4.1.11 and Proposition 2.1.8 that, over geometrically unibranch schemes, constant sheaves of spaces are birational-local.

Example 4.1.14. For any n-birational motivic space $\mathcal{X}$ (see [Mai26d] for the definition), the n-fold $\mathbb{P}^1$-loop space $\Omega^n_{\mathbb{P}^1}\mathcal{X}$ is birational local [Mai26d, Corollary 4.5.2].

Example 4.1.15. Over a perfect field, the infinite loop space of a connective motivic $(S^1/\mathbb{P}^1)$ spectrum is birational-local if and only if the spectrum is zero-sliced (this is the main result of [Mai26a]).

Example 4.1.16 (Sifted colimits of birational local spaces)**.** By Theorem 3.1.1, geometric realizations and filtered colimits of birational local spaces are birational local. For example, an Ind abelian variety is birational.

Example 4.1.17 (through base change)**.** Flat pushforward and smooth pullback of birational local objects are birational local [Mai26d].

4.2. **Examples of Birational equivalences.** The aim of this subsection is to list examples of birational equivalences. Remember that every such morphism is automatically a birational motivic equivalence (see Corollary 2.2.12), so we may use these terms synonymously. To emphasize the difference, we will call a birational morphism of schemes a birational isomorphism. Unless otherwise specified, we will assume that the base is a Qcqs scheme.

Example 4.2.1. Composition, finite product (see Theorem 3.2.2), and colimit of birational equivalences are birational equivalences.

Example 4.2.2. Nisnevich equivalence (Theorem 2.1.3) and cdp equivalences, or more generally cdh equivalences (Remark 2.1.5), are birational equivalences.

Example 4.2.3. Any rational (respectively, retract rational, stably rational, or ordinary) -motivic equivalence is a birational equivalence (see Remark 2.2.11).

Example 4.2.4. A $\mathbb{P}^1$-motivic equivalence in the sense of [Mai26b] is a birational motivic equivalence by Lemma 2.2.8 (1).

Proposition 4.2.5. *Suppose $f : \mathcal{X} \to \mathcal{Y}$ is a morphism of S-spaces. Then f is a birational motivic equivalence if and only if there exists a dense open $V \subset S$ such that f_V is a birational motivic equivalence in $\mathcal{P}(V)$.*

Proof. This is a special case of [Mai26d, Corollary 3.1.9]. □

Example 4.2.6 (through base change)**.** Flat pullback (f^*), smooth extension $p_\sharp$ of birational equivalences, and pullback of an n-birational motivic equivalence along an equidimensional closed immersion are birational equivalences [Mai26d].

Remark 4.2.7. Pullback (in particular, a fiber) of a birational equivalence need not be a birational equivalence. For example, let $0 : \operatorname{Spec} k \to \mathbb{A}^1_k$ be the zero section. Then the pullback of the birational equivalence $\mathbb{G}_m = \mathbb{A}^1_k \setminus 0 \hookrightarrow \mathbb{A}^1_k$ along 0 yields the inclusion $\emptyset \hookrightarrow \operatorname{Spec} k$, which cannot be a birational equivalence, since both $\emptyset$ and $\operatorname{Spec} k$ are birational local over $\operatorname{Spec} k$.

Proposition 4.2.8. *A Nisnevich-local birational motivic equivalence is a birational motivic equivalence. In detail, suppose (S is Qcqs of finite Krull dimension and) $\mathcal{X}_\bullet \to \mathcal{X}$ is a Nisnevich (hypercover/) equivalence of S-spaces, and $f : \mathcal{Y} \to \mathcal{X}$ is a morphism of S-spaces. Then f is a birational equivalence over S if and only if $f_\bullet : \mathcal{Y} \times_{\mathcal{X}} \mathcal{X}_\bullet \to \mathcal{X}_\bullet$ is a simplicial birational equivalence over S.*

Proof. This follows formally from the fact that Nisnevich equivalences are birational equivalences. □

Example 4.2.9. For example, a Nisnevich-locally trivial fibration with a rational group scheme as fiber is a birational motivic equivalence. For example, a GL_n-torsor or an SL_n-torsor is a birational motivic equivalence. On the other hand, since vector bundles (resp. projective bundles) admit Zariski-local trivializations with typical fibers $\mathbb{A}^n$ (resp. $\mathbb{P}^n$), their structure morphisms are birational equivalences.

Proposition 4.2.10. *Suppose $\{p_i : U_i \to S\}_{i\in I}$ is a Nisnevich cover of a Qcqs scheme. Then a morphism of S-spaces $\mathcal{X} \to \mathcal{Y}$ is a birational motivic equivalence over S if and only if, for every $i \in I$, the morphism $\mathcal{X}_{U_i} \to \mathcal{Y}_{U_i}$ of U_i-spaces is a birational motivic equivalence over U_i.*

Proof. The is [Mai26d, Proposition 4.1.1 (1)]. □

Corollary 4.2.11. *Suppose S is a Qcqs scheme and $f : \mathcal{X} \to \mathcal{Y}$ is a 0-birational equivalence of S-spaces. Then for every generic point $\eta \in S^{(0)}$, the map $\mathcal{X}_{k(\eta)} \to \mathcal{Y}_{k(\eta)}$ is birational equivalence over $k(\eta)$. The converse holds when S has finitely many generic points.*

Proof. The first statement follows from [Mai26d, Corollary 3.1.7]. In detail, this follows from the fact that the inclusion of a generic point is a flat morphism and that flat morphisms preserve birational weak equivalences under pullback [Mai26d, Corollary 3.1.6]. The converse is a consequence of [Mai26d, Corollary 4.3.7]. □

Proposition 4.2.12. *Let k be a perfect field and let $f : \mathcal{A} \to \mathcal{B}$ be a morphism of connective S^1-spectra (or, $\mathbb{P}^1$-spectra) such that $s_0 f$ (see [Voe02] for the notation) is an equivalence. Then $\Omega^\infty_{S^1} f$ (resp. $\Omega^\infty_{\mathbb{P}}$) is a birational motivic equivalence.*

Proof. This is a consequence of the Birational motivic recognition principle, to be established in [Mai26a, Corollary 3.5.1]. □

Proposition 4.2.13. *Let k be a perfect field. Then the saturation class $bir(k)$ of birational motivic equivalences is generated in $\mathcal{P}_\Sigma(k)$ by the set of dense open immersions whose closed complement is smooth and has a trivialization of its normal bundle in the ambient scheme.*

Proof. This is immediate from Remark 3.3.8. □

We shall now present a non-example which could otherwise be thought of as a birational equivalence. Namely,

Counterexample 4.2.14. In general, dense open immersions of singular schemes need not be birational equivalences.

Proof. Let k be a field admitting a resolution of singularities (so that every finite-type k-scheme has a cdh cover by a smooth k-scheme). Let A be the union of the two coordinate axes in the affine plane: $A = Spec(k[x,y]/xy)$, and let X and Y denote the two coordinate axes intersecting at the origin, which we identify with $Speck$. Because this gives a cdh square, working in the cdh motivic homotopy category $\mathcal{H}^{\mathbb{A}^1}_{cdh}(Sch_k)$, we find that $L_{cdh,mot}X_+ \simeq \Sigma Speck_+ \simeq S^1$. By [Voe10, Theorem 4.2], this implies that the canonical map $S^2 = \Sigma S^1 \to \Sigma X_+$ is a motivic equivalence in $\mathcal{P}(Sm_k)_\bullet$. Now, the complement of the origin $X \setminus (0,0) \hookrightarrow X$ is a dense open immersion into a singular scheme. So if all dense open immersions in Sch_k are already inverted in $\mathcal{H}^b(k)$, then $S^2 \to \Sigma(X \setminus (0,0))_+$ would be a birational equivalence. But $X \setminus (0,0) \cong \mathbb{G}_m \sqcup \mathbb{G}_m$ is birationally equivalent to $Speck \sqcup Speck$ (i.e., is an equivalence in $\mathcal{H}^b(k)$). Hence, if all (even singular) dense open immersions are already inverted in $\mathcal{H}^b(k)$, then we should have an equivalence $S^2 \to \Sigma(Speck \sqcup Speck)_+ \simeq S^1 \vee S^1$ in $\mathcal{H}^b(k)_\bullet$. But this is impossible, since there are nontrivial commutative birational groups. Indeed, let V be a nontrivial abelian variety over k (which is obviously a commutative monoid that is also birational local by Example 4.1.3). We know that B^2V is birational local by Corollary 3.2.4. So the above equivalence would imply

$$* \cong \mathrm{Map}(S^1 \vee S^1, \mathrm{B}^2V) \simeq \mathrm{Map}(S^2, \mathrm{B}^2V) \simeq V,$$

which is absurd, as V was assumed to be nontrivial. □

Birationally contractible spaces

Definition 4.2.15. *We say that an S-space $\mathcal{Y}$ is birationally contractible if the projection map $\mathcal{Y} \to h_S(S)$ is a birational equivalence. In other words, $L_{bir}\mathcal{Y}$ is the contractible space.*

Example 4.2.16. Every smooth rational scheme is birationally contractible. For example, $\mathbb{A}^n, \mathbb{P}^n, GL_n, PGL_n, Gr_{m,n}, SL_n$ etc. are birationally contractible. More generally, for smooth schemes,

$$\text{rational} \implies \text{stably-rational} \implies \text{retract-rational} \implies \text{birationally contractible}$$

Example 4.2.17. A blowup of a birationally contractible smooth scheme along a smooth center is birationally contractible. Similarly, a vector bundle or a projective bundle over a birationally contractible scheme is birationally contractible (use Example 4.2.9).

Example 4.2.18. Small colimits of birationally contractible spaces are birationally contractible. For example, using Example 4.2.16, we see that $GL, Gr_\infty, \mathbb{A}^\infty, \mathbb{P}^\infty$ are all birationally contractible.

Proposition 4.2.19. *Classifying spaces of birationally contractible monoids are themselves birationally contractible.*

Proof. This is an immediate consequence of Lemma 3.2.3 (2). Indeed, let $\mathcal{M}$ be a birationally contractible monoid. Then,

$$L_{bir}(\mathrm{B}_{nis}\mathcal{M}) \simeq \mathrm{B}(L_{bir}\mathcal{M}) \simeq \mathrm{B}(*) \simeq *.$$

□

Example 4.2.20. For example, $\mathrm{B}_{nis}GL$ is birationally contractible.

Example 4.2.21. Cofibers of birational equivalences (see the examples below) are birationally contractible.

Proposition 4.2.22. *A Severi-Brauer variety over a field k is $\mathbb{A}^1$-connected if and only if it is birationally connected, if and only if it is birationally contractible, if and only if it is isomorphic to $\mathbb{P}^n_k$.*

Proof. Clearly, $\mathbb{P}^n_k$ is birationally contractible, and a birationally contractible space is birationally connected. The identification of birational connectivity with $\mathbb{A}^1$-connectivity is Corollary 3.6.22.

Let X be a Severi-Brauer variety over k such that $\pi^b_0 X \cong *$. Since X is proper, by Theorem 3.6.15

$$X(k)/_\sim \cong \pi^b_0 X(k) \cong *,$$

so $X(k) \neq \emptyset$. Since X is a Severi-Brauer variety, this implies $X \cong \mathbb{P}^n_k$ [GS17, Theorem 5.1.3]. □

Corollary 4.2.23. *A smooth, proper curve over k is birationally connected if and only if it is $\mathbb{A}^1$-connected if and only if it is birationally contractible if and only if it is isomorphic to $\mathbb{P}^1_k$.*

Proof. The proof is identical to the one above, using the fact that such curves must be of genus 0 (see Corollary 4.1.6) and that $\mathbb{P}^1_k$ is the unique smooth proper curve of genus 0 with a k-rational point. □

Remark 4.2.24. A similar characterization of smooth, proper, birationally contractible surfaces does not hold, unlike the $\mathbb{A}^1$-local characterization (over fields of characteristic zero) of the affine plane as the unique smooth $\mathbb{A}^1$-contractible surface [DVO26, Theorem 3.12]. Indeed, both $\mathbb{P}^1_k \times \mathbb{P}^1_k$ and $\mathbb{P}^2_k$ are smooth, proper, birationally contractible surfaces. In fact, by Example 4.2.17, every blow-up of either is a proper, birationally contractible surface. There are many other proper, birationally contractible surfaces beyond these. For example, for every $n \geq 0$, the Hirzebruch surfaces

$$\Sigma_n := \mathbb{P}_{\mathbb{P}^1}(\mathcal{O}_{\mathbb{P}^1}(-n) \oplus \mathcal{O}_{\mathbb{P}^1})$$

are $\mathbb{P}^1$-bundles over the birationally contractible space $\mathbb{P}^1$ and are therefore birationally contractible (use Example 4.2.17 again). (Note that the Hirzebruch surfaces are in fact $\mathbb{P}^1$-locally contractible in the Nisnevich topology [Mai26b, Example 2.2.20].)

Corollary 4.2.25. *Let $\mathcal{X}$ be birationally contractible over S. Then, for every generic point $\eta \in S$, $\mathcal{X}_{k(\eta)}$ is birationally contractible over $k(\eta)$.*

Proof. This is an application of Corollary 4.2.11. □

More generally,

Proposition 4.2.26. *Let $f : \mathcal{X} \to S$ be an n-birational motivic equivalence (see the paragraph below* [Mai26d, Definition 2.2.1.]*) over a noetherian, universally catenary scheme S. Then for every $s \in S^{(n)}$, the fiber $\mathcal{X}_{k(s)}$ is birationally contractible over $k(s)$.*

Proof. This is immediate from [Mai26d, Corollary 3.2.3]. □

Birational motivic equivalence of schemes

Example 4.2.27. A birational morphism, or, more generally, a stably birational morphism of smooth schemes, is a birational equivalence (since $\mathbb{P}^n$ is birationally contractible and taking products with a scheme preserves birational equivalences, Theorem 3.2.2).

Remark 4.2.28. However, it remains unclear whether birational isomorphisms (or even dense open immersions) of non-smooth schemes are birational equivalences, since we have only inverted the dense open maps between smooth schemes (see also Remark 3.1.8). In contrast, vector bundle projections over arbitrary (not necessarily smooth) X/S are motivic equivalences over S (use the fact that L_{mot} is cartesian and that Zariski (or Nisnevich) hypercovers of such arbitrary X's are Nisnevich equivalences in $\mathcal{P}(S)$).

Proposition 4.2.29. *Over a field, a birational equivalence between 0-dimensional schemes is an isomorphism.*

Proof. By Example 4.1.2, such schemes are birationally rigid and hence birational local. Thus, a birational local equivalence between such spaces is an equivalence. But an equivalence of discrete spaces is an isomorphism. □

Proposition 4.2.30. *Over a field, a birational equivalence between proper curves of positive genus is an isomorphism.*

Proof. By Proposition 4.1.5, it follows that such curves are birationally rigid. Thus, the claim follows similarly to the proof above. □

Example 4.2.31. Over an algebraically closed field, a morphism of proper curves is a birational equivalence if and only if it is an isomorphism. The point is that, in this case, curves of genus 0 are all isomorphic to $\mathbb{P}^1_k$ (by the existence of a rational point), which is birationally contractible.

Proposition 4.2.32. *Suppose $f : X \to Y$ is a morphism of smooth S-schemes. Then f is a birational motivic equivalence if and only if there exists a dense open $V \subset S$ such that f_V is a birational motivic equivalence in $\mathcal{P}(S)$.*

Proof. This follows from the 2-out-of-3 property of birational motivic equivalences and the stability of dense open immersions under smooth pullback (see Lemma 2.1.2). □

Proposition 4.2.33. *Suppose $f : X \to Y$ is a morphism of smooth schemes over S with n-birationally contractible fibers over an n-dense open $V \subset Y$, i.e., $f_V : X_V \to V$ is an n-birational motivic equivalence in $\mathcal{P}(V)$. Then f is an n-birational equivalence in $\mathcal{P}(S)$.*

Proof. This is [Mai26d, Corollary 3.1.22]. □

Proposition 4.2.34. *Suppose S is a Qcqs scheme with finitely many irreducible components and $f : X \to Y$ is a morphism in Sm_S such that the generic fibers of f are birationally contractible, i.e., for every generic point $\eta \in Y^{(0)}$ the scheme $X_{k(\eta)}$ is birationally contractible over $k(\eta)$. Then f is a birational equivalence in $\mathcal{P}(S)$.*

Proof. See [Mai26d, Corollary 4.3.6]. □

Corollary 4.2.35. *Suppose $f : X \to Y$ is a morphism of smooth schemes over a field k such that the fiber of f over every generic point $\eta \in Y^{(0)}$ is a smooth, retract-rational scheme over $k(\eta)$. Then f is a birational motivic equivalence.*

Remark 4.2.36. When the fibers are rational, it is easy to see that the original morphism is a stably birational isomorphism of schemes, which is a rational motivic equivalence and hence a birational motivic equivalence, Example 4.2.27. However, a morphism with retract rational fibers might not, in general, be a retract rational motivic equivalence (see Remark 2.2.11 for the definition) (the problem is similar to the remark below), let alone a stably birational isomorphism.

Remark 4.2.37. Because the birational homotopy category is less well-behaved under base change than the motivic homotopy category (specifically, closed immersions may not preserve all birational weak equivalences under pushforward; see [Mai26d]), the techniques of [DVO26,

Proposition 1.15] (which apply to the ordinary motivic case for arbitrary Qcqs schemes (say, of finite Krull dimension)) cannot be applied to the birational motivic localization to deduce Proposition 4.2.34. This is well known to be possible for motivic equivalences. In this section, we want to single out a larger saturated class. Unfortunately, even rational schemes fail to be stable under arbitrary base change, suggesting that this required property may even fail for the saturated class of rational equivalences.

Étale birational equivalences

We now briefly discuss a way to characterize the generating set of birational (isomorphisms and) equivalences, namely the dense open immersions among birational isomorphisms and birational equivalences.

The geometric characterization of dense open immersions is well known. We recall it here for the sake of completeness and to generalize it to a homotopical statement:

Lemma 4.2.38. *A finite étale birational map of schemes is an isomorphism.*

Proof. Since f is finite étale, the structure map $\mathcal{O}_Y \to f_*\mathcal{O}_X$ is a locally free module of constant rank on each connected component. Since this is an isomorphism over each generic point of Y, it is globally of rank 1 and hence is an isomorphism. Since f is affine, so that $X \simeq Spec_{\mathcal{O}_X}(f_*\mathcal{O}_Y)$, the claim follows. □

Lemma 4.2.39. *An étale morphism of qcqs schemes is a birational isomorphism if and only if it is a dense open immersion.*

Proof. The 'if' part is clear.

Let $f : X \to Y$ be an etale map. Since openness is a local condition, we may assume that f is separated. Since f is quasifinite, by Zariski's main theorem [Sta26, Lemma 05K0] it factors as a finite map $p : W \to Y$ followed by an open immersion j. By taking the closure of the image of j (with the reduced scheme structure), we may assume that j is a dense open immersion and p is finite. By the two-out-of-three property for etale and birational maps, the finite map p is also etale and birational. But then the lemma above implies that p is an isomorphism. Thus $f \cong j$, which is a dense open immersion. □

We now prove the homotopical statement:

Corollary 4.2.40. *Suppose S is Qcqs and $f : X \to S$ is any morphism. Then the following are equivalent:*

(1) f is a dense open immersion.
(2) f is étale and a birational equivalence over S.
(3) f is etale and has birationally contractible generic fiber over each generic point of S.

Proof. Since open immersions are étale, (1) $\implies$ (2) follows from definition of birational equivalence. (2) $\implies$ (3) is immediate from Corollary 4.2.11.

Now suppose $f : X \to S$ has birationally contractible generic fibers. If f is etale, then the generic fibers $X_{k(\eta)}$ (in fact, all fibers) are zero-dimensional and hence birationally rigid over $k(\eta)$ (Example 4.1.2). Since $f_\eta : X_{k(\eta)} \to Speck(\eta)$ is given to be a birational equivalence, it follows that it is actually an equivalence and thus an isomorphism (as the domain and codomain are discrete). Since our morphisms are of finite presentation, these isomorphisms can be extended to a

dense open subscheme of S. This way, f becomes a birational isomorphism. From Lemma 4.2.39, we then know that f is a dense open immersion. The equivalence of the last two statements (even without the étale condition) follows from Corollary 4.2.11. The claim then follows from the theorem above. □

Theorem 4.2.41. *Let S be a Qcqs scheme. Then $f : X \to Y$ in Sm_S is a dense open immersion if and only if it is etale and has a birationally contractible generic fiber over each generic point of its codomain.*

Proof. Suppose $p : Y \to S$ is the smooth structure map of Y. Since $p_\sharp$ preserves birational equivalences (Example 4.2.6), we may reduce to the case $Y = S$, where this is just the equivalence of conditions (1) and (3) in Corollary 4.2.40. □

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